

Intelligent cryptocurrency trading system using integrated AdaBoost-LSTM with market turbulence knowledge

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ABSTRACT

The Bitcoin market is firmly positioned as a global asset market. However, due to its extremely high volatility and the lack of a custom Bitcoin trading system, investors find it difficult to establish an effective investment strategy in this market. In this study, we build an intelligent Bitcoin trading system to maximize profitability by predicting the market. We propose a fusion approach combining technology with economic knowledge to achieve accurate predictions. Firstly, we provide an ensemble prediction framework that integrates the AdaBoost algorithm with the LSTM deep learning model as a form of technological convergence. This dramatically improves the predictive performance due to the high representational capacity of LSTM and the overfitting minimization capability of AdaBoost. Secondly, to account for the repeated structural volatility in the market, we combine the econometrics Markov regime-switching model with the AdaBoost-LSTM model to predict the period of market turbulence caused by structural breaks in the regime. Our prediction model suggests that market turbulence lasting longer than 21 days could be a potentially high-risk investment in the future. Finally, we have developed a Bitcoin-customized trading algorithm that maximizes rewards by predicting upward movements in prices and warning of high-risk investments during a long-lasting turbulent regime. In trading tests, our system yielded a cumulative return of 2.4 times and 4.0 times on 1-day and 5-day forecasts, respectively, compared to other representative trading strategies such as the stochastic oscillator.

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1. Introduction

The cryptocurrency market¹ has been booming in recent years. Bitcoin is the most successful decentralized cryptocurrency system. While Bitcoin cannot be used to make payments or generate dividends due to its lack of intrinsic value, it has been recognized as an investment asset that can be sold at a higher price when

the supply of digital currency is limited. For example, the onset of the COVID-19 pandemic stimulated public demand for cryptocurrency to hedge against high inflation caused by massive amounts of liquidity [1,2].

However, Bitcoin has become well known for its extremely high volatility, several dozen times higher than that of other asset markets such as stocks, gold, and currency [3,4]. Accordingly, the Bitcoin market is characterized by high risk, high rewards, repeated patterns of unusual returns, and frequent falls. Many economists have argued that well-known trading strategies such as moving average convergence divergence (MACD), rate-of-change (ROC), and the relative strength index (RSI) are not useful for Bitcoin investment, and there is still no cryptocurrency trading system that guarantees stable investment [5,6]. Since March 2022, the tightening of U.S. monetary policy has increased uncertainty and volatility in the cryptocurrency market, which has threatened the popularity of cryptocurrency investment, with Bitcoin prices falling by −53.4% compared to the beginning of the year. Even trading volume is on the decline. Because the cryptocurrency market is now established as a global asset market, our study addresses the need for a Bitcoin trading system similar to those for other assets in order to reestablish popularity and ensure the reliability of cryptocurrency investment for investors.

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¹ As of August 20, 2022, the Bitcoin market cap accounted for 40.0% (\$405.7 billion) of the total market, which consists of 20,652 cryptocurrencies, 512 exchanges, and a market size of approximately \$1.02 trillion. Source: CoinMarketCap.

The most important thing to build an efficient and stable cryptocurrency trading system is accurate prediction despite the high volatility and potentially high risk in this market. Initially, linear relationship-based statistical models [3,7,8] were used to predict Bitcoin prices, but prediction research has recently shifted towards machine learning (ML) [9–11] to analyze the complex, non-linear relationships between various factors and Bitcoin markets. However, it is difficult to extract features that can help predict the interrelationships between variables using traditional ML models such as artificial neural networks (ANNs), support vector machines (SVM), and random forest (RF) due to their shallow learning structures [12].

Recently, the DL model has been utilized to predict complex patterns in the Bitcoin market due to its high representational capacity based on large samples even in the absence of detailed assumptions about learning [13]. Research on DL for Bitcoin market prediction has generally been bifurcated into studies using convolutional neural networks (CNNs) [14] and those using long short-term memory (LSTM) networks [11,15]. While CNNs can reduce high dimensionality without loss of information, LSTM has proven excellent for learning long-range patterns due to its use of recurrent and gate mechanisms. In addition, the LSTM model is being used more frequently for time-series prediction.

However, LSTM requires a large number of parameters to handle long-term dependencies, especially in higher dimensions, where the number of parameters dramatically increases, leading to overfitting problems and loss of predictive power [16]. Many inputs such as financial and macro-level variables are required to deal with the high volatility and uncertainty of the Bitcoin market. In addition, heterogeneous inefficiencies may be observed in the Bitcoin market because of a lack of regulations compared to other financial assets [17,18]. Since such inefficiencies often cause turbulent volatility in the Bitcoin market [19,20], forecasting techniques alone cannot address this challenge. For example, after the bull market at the beginning of 2017, Bitcoin prices hit a historical peak on December 18th, 2017, which was followed by a crash due in which investors suffered a loss of 80% the following year.

Many economists [21,22] have argued that market turbulence may be characterized by explosive fluctuations and long-term persistence caused by structural breaks in the market. To analyze market turbulence, the non-linear-based Markov regime-switching (MS) proposed by Hamilton et al. (1990, 1994) [23,24] has been widely used in economics research. The MS model divides the market into either tranquil or turbulent regimes, focusing on the possibility of a structural break in mean returns and variance of residuals. It has the advantage of probabilistically predicting prices in a dynamically changing market; it has been utilized for bubble detection and risk warning in traditional assets markets [25–27] as well as for cryptocurrencies [28,29]. However, the MS model is limited in focusing only on time-series price fluctuations and not considering the effects of various other macroeconomic factors and blockchain technology.

Accordingly, the primary goal of our study is to increase the predictability and profitability of Bitcoin trading, as follows: First, we design a stable DL framework in which overfitting is minimized even with numerous functions related to the Bitcoin market. Second, we develop an intelligent trading algorithm to avoid high risk and increase returns by considering market turbulence. To achieve our objective, we employ a fusion strategy that enhances the prediction power based on combinational approach that can aid each other by integrating data or characteristics. In particular, we propose a fusion approach that expands sequentially in three stages: model fusion, information fusion, and feature fusion. Model fusion is the integration of heterogeneous

forecast models, and information fusion includes merging or expanding data from multiple sources. The feature fusion reveals a potential inherent feature from a combination of partial features.

First, we suggest the AdaBoost-LSTM methodology that integrates an ensemble framework with LSTM models. Ensemble learning can reduce high variance due to overfitting by combining a number of weak models that are slightly better than random guessing, and it can fundamentally solve the overfitting problem without the loss of information even with a hundred features [30]. We fuse the LSTM model with the adaptive boosting (AdaBoost) framework proposed by Freund and Schapire (1997) [31]. AdaBoost can reduce overfitting and improve the predictive ability in the combined model to achieve sequential ensemble learning adaptively. We hypothesize that using the ensemble model alone will still perform less well than the LSTM model due to the latter's superior representational capacity for complexity, but combining the two can improve the prediction accuracy of the LSTM.

Second, we capture the turbulence information of the Bitcoin market using the MS model to investigate the influence of high-risk turbulence on Bitcoin price over time. By combining Markov modeling of market turbulence with our AdaBoost-LSTM model and analyzing relational patterns between volatility and various features such as macroeconomic and blockchain technology, we achieve a more accurate prediction of price movement and turbulence in the future market.

Finally, based on the combination of prediction results, we implement a trading system to generate maximum profit in the Bitcoin market. In particular, our trading system considers a prolonged regime of volatility as a market turbulence feature associated with high risk. The practicality of our trading system is compared with traditional trading strategies such as MACD, ROC, and the stochastic oscillator.

Overall, the expected contributions of our study are threefold:

1. We propose a novel prediction method that integrates the AdaBoost ensemble framework with LSTM to prevent overfitting even with many features. Our model fusion approach, which complements the shortcomings and strengths of each framework, differs from those in previous studies that utilize prediction techniques independently.
2. We predict turbulence in the Bitcoin market with consideration of changes in the global financial system, which is useful knowledge to avoid high-risk investments.
3. We design an intelligent trading system that maximizes profits by predicting upward movement in the Bitcoin market and warning against trading during high-risk periods of prolonged turbulence.

The remainder of the paper is structured as follows. Section 2 reviews the technology and literature related to this work. Section 3 provides details of our proposed regime detection and ensemble prediction methodology. Section 4 outlines the results of the empirical analysis, presenting evidence of market turbulence and describing the performance of our predictive model and Bitcoin trading system. Finally, some concluding remarks and limitations are provided in Section 5.

2. Related works

2.1. Market turbulence

Many previous studies have focused on mechanisms of transition from turbulent regimes in the international asset market, including stocks [27], exchange rates [32], real estate [33], and cryptocurrency [28,34]. Identifying turbulence in the asset market is essential for diversifying portfolios and avoiding prolonged

volatility, which may cause a crisis. Many studies [21,27,32] have demonstrated a close correlation between extreme volatility and several crises such as the global financial crisis of 2008 and the Mexican and Asian crises of the 1990s.

Many economists have derived the characteristics of market turbulence using econometric techniques such as Value-at-Risk (VaR) [35], autoregressive integrated moving average (ARIMA) [36], and generalized autoregressive conditional heteroskedasticity (GARCH) [34] on the premise that asset price volatility has time-series dependence. However, the above models are based on regression analysis, which is limited in capturing explosive conditional variance caused by persistent volatility in asset markets [21]. In general, excessive turbulence occurs due to structural breaks [22]. Hamilton et al. (1990, 1994) [23,24] proposed the non-linear MS model that identifies unobserved persistent regimes (turbulent or tranquil) by calculating the possibility of a structural break based on the mean value of asset returns and variance of the error term in the autoregressive model. Because MS using Markov chain transition probabilities can adequately capture the dynamics of a volatile market, it has been popular in studies of various asset markets such as stocks [27], oil [25], gold [26], and currency [32] as well as cryptocurrency [28,29]. In particular, the two-state MS model has been used to implement risk warnings [27,28], detect asset price bubbles [26,37], and optimize portfolios [29]. Hybrid versions such as MS-GARCH [38] and MS-VAR [39] have been combined with various time-series models, but the classical MS model remains the most popular.

However, the MS model focuses only on price fluctuations and does not consider that changes in the regime in cryptocurrency markets can also be caused by macroeconomic factors and blockchain technology conditions. Recently, changes in macroeconomic factors and the global financial environment were observed due to COVID-19, as a result of which uncertainty in the cryptocurrency market has continued to increase [40]. Therefore, it is necessary to consider the influence of the financial system on changes in regime in order to capture sufficient information in an excessively turbulent market.

2.2. Bitcoin trading system

Existing trading systems predict the direction of financial markets by analyzing historical data such as asset prices and volume. Unlike traditional asset markets such as stocks and foreign exchanges, the cryptocurrency market is relatively new; therefore, there are few studies on trading strategies for investment in this market. Research on the cryptocurrency trading system mainly uses technical indicators originally applied in the traditional asset market. Representative technical indicators such as MACD, ROC, RSI, and the stochastic oscillator have been used to forecast price movements and provide a “buy” signal to investors [5,6,41]. However, due to the volatility of cryptocurrency, there is a limit to the extent to which market risk can be predicted using a technical indicator, and, in severe cases, predicting profitability is worse than recommending buying and holding [5].

Recently, trading researchers attempted to predict the direction of the cryptocurrency market based on AI and ML. Serrano (2022) [42] used the addition of deep learning to a reinforcement learning algorithm to predict the upward and downward of Bitcoin prices. In addition, Borges and Neves (2020) [43] created four new technical indicators using trading indicators as inputs in several ML models. Further, Ortu et al. (2022) [44] improved the power of their model to predict the direction of the cryptocurrency market by combining a trading indicator and a social media indicator.

2.3. Bitcoin price prediction

Many studies have focused on predicting Bitcoin prices. Their approaches can be broadly divided into two categories [45]: statistical models and ML models. Statistical models such as ARIMA, GARCH, and exponentially weighted moving average (EWMA) have been widely used in Bitcoin prediction [3,7,8]. However, these models can only handle linear problems based on normal distribution, and the use of many variables leads to multicollinearity problems [46,47].

ML techniques such as ANNs [48], SVM [49], and RF [9] have also been utilized for predicting Bitcoin prices because these models can analyze complex non-linear relationships among multiple variables [50,51]. DL approaches have shown excellent performance for financial time-series forecasting due to their high representational capacity [52]. Among DL models, the LSTM model has outperformed other ML models by learning long-range patterns better using recurrent and gate mechanisms [11,15]. However, as the number of input variables increases, the DL model requires many parameters to handle such high dimensionality, which may cause overfitting problems [16]. Cho et al. (2014) proposed GRU as a simplified version of LSTM that reduces the parameters in the gate mechanism. GRU has a faster learning time and is more stable in overfitting, but it still requires more parameters than some ML models.

Recently, some studies combined DL with dimensional reduction techniques such as principal component analysis and wavelet transform [53], but these reduction techniques can lead to loss of information and are sensitive to outliers, which can lead to errors [54].

In general, models overfitted by training sets have high variance and slight bias. The ensemble learning framework reduces bias and/or variance by combining several low-biased weak models, which are slightly better than random guesses [55]. Recently, ensembles have been essential for ML pipelines, demonstrating robust and reliable feature selection in high-dimensional data [56]. In particular, Ganaie et al. (2022) [57] showed several deep ensemble learning strategies to combine the DL model and ensemble learning methods.

Two approaches to representative ensemble learning are bagging and boosting [58]. The concept of bagging, proposed by Breiman (1996) [59], is that weak models learn independently in parallel and may be combined following a deterministic averaging process. A representative bagging model is RF, which has been frequently used in cryptocurrency prediction [9,60]. By contrast, the boosting concept proposed by Schapire (1990) [61] is that weak models learn sequentially and adaptively (i.e., the current model is based on the previous model). Adaptive boosting (AdaBoost) [31] and extreme gradient boosting (XGB) [62] are representative boosting models. The boosting-based prediction has also been actively applied in the context of cryptocurrency [9,10].

Boosting algorithms usually outperform bagging algorithms, except for data characterized by noise (i.e., data sets with incorrect class labels and imbalanced class ratios) [63,64]. AdaBoost has been particularly popular because it has the following advantages: it sequentially improves the predictive power of ensemble models using weights measuring the prediction errors generated by each model, and it reduces overfitting [16]. Although the AdaBoost algorithm is highly scalable due to its simple mechanism and requires few parameters, its exponential loss function can increase the prediction error by overestimating the noise data, such as outliers [65].

XGBoost developed by Chen et al. (2016) [62] is robust against noise data by using the gradient descent loss function. In addition, the learning speed was improved by the parallel tree boosting method.

Recent studies have recommended the combination of an ensemble framework with other techniques such as ML models to achieve greater success in real applications [57,58]. In particular, combining AdaBoost and the LSTM model has especially proven to improve the accuracy of prediction in asset markets such as the raw materials market [66], stocks [67], manufacturing [68], and energy [69]. In Bitcoin price prediction, Boukher et al. (2022) [70] demonstrated that the prediction performance based on the AdaBoost-LSTM methodology was improved by 19.29% more than that of the LSTM alone model.

However, existing studies using the AdaBoost-LSTM method have limitations in that they focus only on the improvement of the technology itself. Heterogeneous inefficiencies such as herding behavior may be observed in the Bitcoin market because of a lack of regulations compared to other financial assets [17,18]. Since such inefficiencies often cause turbulent volatility in the Bitcoin market [19,20], prediction techniques alone cannot overcome this challenge.

Therefore, knowledge modeling of the inefficiency information of the Bitcoin market and the development of Bitcoin price prediction techniques must be carried out together. To this end, multimodal models for various purposes are required, and the main task is how to fuse these heterogeneous models.

3. Proposed methodology

The following research questions were posed to determine how our AdaBoost-LSTM ensemble prediction method with knowledge of market turbulence through Markov modeling can minimize risk and maximize rewards in the Bitcoin market.

- Q1: How does Markov modeling represent turbulence in the market? How does market turbulence affect the Bitcoin market?
- Q2: How can the ensemble framework be fused with the LSTM model? How much does the ensemble methodology improve predictive performance?
- Q3: How specifically is the Bitcoin trading system built? How profitable is our Bitcoin trading system compared to existing trading strategies such as MACD, ROC, and stochastic oscillator?

To address these questions, our fusing approach is performed with two-phase modeling, as shown in Fig. 1.

In phase I, the turbulent regime knowledge in the Bitcoin market is built using the MS model, and we describe in detail how turbulence affects the Bitcoin market. In phase II, a Bitcoin trading system is built based on prediction. First, we construct the ensemble prediction framework by designing the basic structure of the LSTM model composed of weak learners and the optimal number of LSTMs to be adaptively included. Next, we build predictive models based on the ensemble framework and turbulence knowledge, representing price movement and turbulence in the Bitcoin market. The predictive models are trained using various macro-financial and blockchain variables related to the Bitcoin market. Finally, based on the potential inherent features of the Bitcoin market obtained by integrating the two heterogeneous ensemble prediction models, we implement a Bitcoin trading system that maximizes the reward by avoiding risk.

3.1. Data preparation

Bitcoin was launched in 2009, but the trading boom began in 2016 with a lot of attention from investors [71]. In addition, after the outbreak of COVID-19, there was a “black swan” effect that caused extreme volatility in the Bitcoin market [72].

Our data set² represents the daily set from June 1, 2016 to April 30, 2022, and is divided into a training set (from June 1, 2016 to March 31, 2021) and a test set (from April 1, 2021 to April 30, 2022). We assign 1/10 of the training set as the validation set. Thus, the entire data set is divided into training, validation, and testing periods at a ratio of 0.7:0.1:0.2.

Our training set can learn the uncertainty pattern of the Bitcoin market from Jan to October 2020 during the outbreak of COVID-19, and the trained model is verified through a validation set. Table 1 shows descriptive statistics of logarithmic Bitcoin returns in a train, validation, and test set and verifies the stationarity of the time-series data set using the Dickey–Fuller (ADF) test. The ADF statistics for the daily log return are the train (−25.75), validation (−9.33), and test set (−13.36), respectively, both less than the threshold of −3.46. Therefore, it indicates that the null hypothesis for non-stationarity was rejected at a significance level of 0.01.

In addition, our study used various Bitcoin-related variables and financial variables to predict the Bitcoin market.

Table 2 lists 16 variables related to financial and economic factors, blockchain information, public attention, and market turbulence, all of which have been proven to affect Bitcoin prices in previous research.

This data structure has been commonly used in previous research to build a stable ML model [13,74]. All data was collected from public databases such as FRED (<https://fred.stlouisfed.org/>), Coinmarketcap (<https://coinmarketcap.com/>), and Nasdaq Data Link (<https://data.nasdaq.com/>).

Table 3 shows the summary statistics of all variables used in our analysis.

ML performance is heavily dependent on a variety of input variables [87]. Therefore, we perform min–max normalization to set the scales of all variables X within the range of 0–1, as follows: $\hat{X} = \frac{X - \text{Min}(X)}{\text{Max}(X) - \text{Min}(X)}$, where the $\text{Min}(X)$ and $\text{Max}(X)$ functions respectively indicate the minimum and maximum values of X . For missing values in the data, we implement specific imputation approaches (e.g., average, median).

3.2. Modeling phase I

Turbulence in the asset market is often the inevitable result of volatility jumps. To investigate this turbulence in the Bitcoin market, we employ the MS model developed by Hamilton et al. (1990, 1994) [23,24] to identify structural changes in Bitcoin returns. Our MS model is based on two assumptions: (1) the volatility process has two regimes, turbulent and tranquil, and (2) the turbulent regime is related to large deviations in price returns (= high value from mean processes), and the tranquil regime is related to small fluctuations in returns (= low values from mean processes). Many scholars have argued that these two states best describe the fundamentals of asset markets [28,32,88].

Given Bitcoin return y_t at time t , a two-state switching autoregression with a single lag ($\gamma = 1$) is formulated as:

$$y_t - \mu_{s_t} = \gamma(y_{t-1} - \mu_{s_{t-1}}) + \varepsilon_t, \varepsilon_t \sim i.i.d.N(0, \sigma^2), \quad (1)$$

where s_t is the state variable, $s_t = 1$ indicates the turbulent state, $s_t = 0$ represents the tranquil state, μ_{s_t} is the mean of each state, and the error term ε_t is independent and identically distributed random variables. We label our MS model including a single lag and two regimes as MS(2)-AR(1).

² Bitcoin is traded on both weekends and weekdays, while the daily data for financial and economic variables is collected on weekdays only. Therefore, we utilize values of the previous weekday for financial and economic variables for weekend days, as suggested by Dutta et al. (2020) [73].

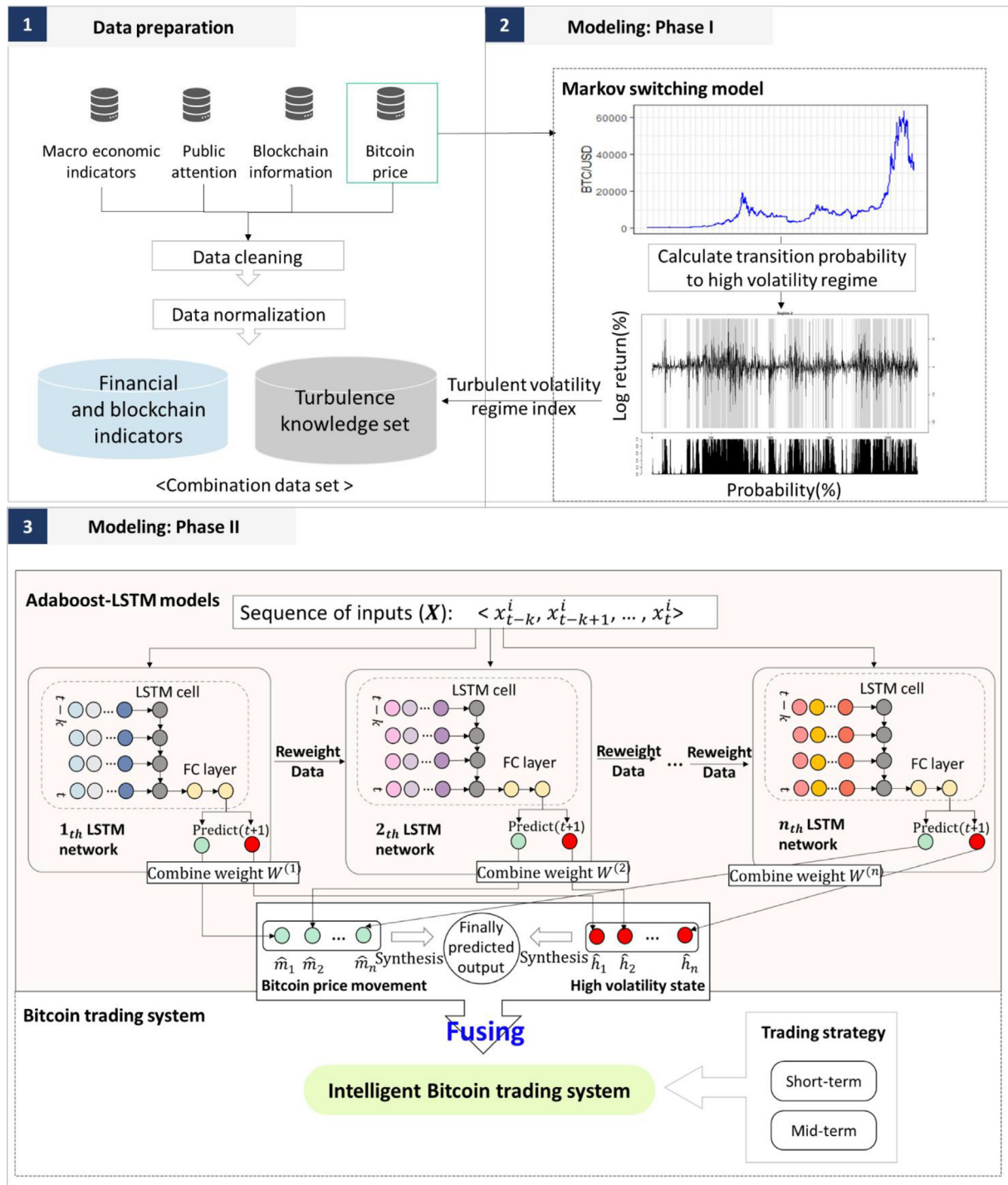


Fig. 1. Two-stage modeling employed in AdaBoost-LSTM prediction and Bitcoin trading system.

Table 1

Descriptive statistics of logarithmic Bitcoin returns in training, validation, and test set.

	Period	N (Ratio)	Logarithmic returns					Price direction		Percentage(%)	
			Mean	Median	Skew	SD	ADF	Up	Down	Up	Down
Entire set	Jun 1, 2016–Apr 3, 2022	2160(1.0)	0.20	0.19	−0.39	3.26	−30.36*	1174	986	54.35	45.65
Train	Jun 1, 2016–Oct 5, 2020	1588(0.7)	0.19	0.17	−0.45	3.25	−25.75*	874	714	55.04	44.96
Validation	Oct 6, 2020–Mar 31, 2021	177(0.1)	0.96	1.01	−0.31	3.63	−9.33*	118	59	66.67	33.33
Test	Apr 1, 2021–Apr 31, 2022	395(0.2)	−0.10	−0.10	−0.35	3.07	−13.36*	182	213	46.08	53.92

* ADF-statistics means that the null hypothesis is rejected at the 1% significance level.

Table 2

Variables related to the daily price of Bitcoin (related literature in square parentheses).

Attribute	Variable	Definition	Literature
Macro-economic and finance	10-year bonds	U.S. 10-year treasury bond yield (%)	[3,4,75]
	2-year bonds	U.S. 2-year treasury bond yield (%)	
	Gold	Price of gold in the Chicago Mercantile Exchange (against the U.S. dollar)	[3,76]
	Nasdaq	U.S. equity index	[77,78]
	S&P 500		
	VIX	Benchmark measure of expected stock market volatility of S&P 500	[79]
	Dollar index	Value index of the U.S. dollar against a basket of foreign currencies	[3]
Public attention	WTI	Price of crude oil for West Texas Intermediate (against the U.S. dollar)	[80,81]
	Tweets	Normalized search volume of the inquiry “Bitcoin” per day	[82,83]
Blockchain technology	Price	Bitcoin closed exchange rate price (against the U.S. dollar)	
	Log return	Logarithmic return of bitcoin trading (%)	
	Block size	Average block size (MB)	[82,84–86]
	Market capital	Total U.S. dollar market value of Bitcoin (in millions)	
	Number of transactions	Number of transactions per day (in millions)	
	Trade volume	Amount of Bitcoin traded per day (in millions)	
	Hash rate	Number of hashes performed by the Bitcoin network (trillions per second)	

Table 3

Summary statistics of daily variables between June 1, 2016 and April 30, 2022.

Variable	N	Mean	SD	Median	Min	Max	Skew	SE
10-year bonds	2160	1.94	0.71	1.92	0.51	3.24	−0.21	0.02
2-year bonds	2160	1.28	0.89	1.31	0.11	2.97	0.16	0.02
Gold	2160	1500.90	262.03	1353.20	1129.80	2069.40	0.44	5.64
S&P 500	2160	9085.32	3221.71	7902.54	4594.44	16057.44	0.70	69.32
Nasdaq	2160	3097.64	749.46	2872.87	2000.54	4796.56	0.76	16.13
VIX	2160	18.27	8.33	16.16	9.14	82.69	2.53	0.18
Dollar index	2160	95.54	3.18	95.88	88.59	103.62	0.00	0.07
WTI	2160	57.43	15.36	55.66	−37.63	123.70	0.61	0.33
Tweets	2160	56122.63	47992.56	34247.00	13294.00	363566.00	1.75	1032.64
Price	2160	15468.84	17446.15	8255.62	533.15	67523.00	1.36	375.38
Log return	2160	0.20	3.26	0.19	−22.15	17.39	−0.39	0.07
Block size	2160	1.05	0.20	1.04	0.43	1.53	−0.17	0.00
Market capital	2160	284752.21	329940.33	145989.22	8421.13	1277657.9	1.37	7099.19
Number of transactions	2160	0.28	0.05	0.28	0.12	0.49	0.02	0.00
Trade volume	2160	698.96	844.58	381.62	7.69	9988.78	2.93	18.17
Hash rate	2160	74.44	62.34	54.19	1.18	248.11	0.46	1.34

The probability law determines the regime at time t with conditional probability depending on what the regime was at previous time $t - 1$, so that state S_t at time t forms a time-homogeneous first-order Markov chain on the two states: $\{0, 1\}$ with a probability of transition, which is explained by:

$$P_{ij} = \text{Prob}(S_t = j | S_{t-1} = i), i, j = \{0, 1\}. \quad (2)$$

The two-state regime can be classified according to filtering probability, which is the maximization of the log-likelihood estimation. The filtering probability based on historical observations till time t , $Y_t = (y_1, y_2, \dots, y_t)$ is written as:

$$P(S_t = i | Y_t; \theta_t), \quad (3)$$

where θ_t is the vector of model parameters to be estimated. $P(S_t = 1 | Y_t; \theta_t)$ can be interpreted as the conditional probability of a turbulent state based on the current information at time t .

In this study, we define a turbulent regime as occurring when the probability of a transition to high volatility exceeds a certain threshold λ , which is described by the following binomial function:

$$\text{Turbulent} - \text{regime period} = \begin{cases} 1, & P(S_t = 1 | Y_t; \theta_t) > \lambda \\ 0, & \text{otherwise}, \end{cases} \quad (4)$$

where λ represents the lowest likelihood of high volatility that can exist and still be considered a turbulent regime. Therefore, determining λ is essential. Many studies [27,89] have used a

two-peak method that determines the optimal threshold for a binary system as the minimum value between two peaks of the frequency density histogram; our study adopts the same criterion. Specifically, we draw a histogram of filtering probability for the turbulent state and select the bottom of the valley between the two frequency peaks as the cutoff point. Finally, the turbulent state dummy is set to 1 for a period with a probability exceeding the cutoff point, and the rest is set to 0 for a tranquil period otherwise. We include our dummy index and probability values related to the regime as variables in our ML model.

3.3. Modeling phase II

In modeling state II, we propose an ensemble framework for the prediction of Bitcoin prices and classifying of turbulent regimes, and we develop a profitable trading system based on the results. To achieve our objective, first, we build the foundation of our ensemble model based on DL techniques with superior representational capacity for high-dimensional and complex patterns. In particular, we employ LSTM DL because of its advantage of learning long-range patterns [90].

In order to evaluate the model's prediction performance, we use different performance indicators for prediction purposes, such as regression problems and classification problems.

(1) Regression problem:

$$(1) \text{ RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2},$$

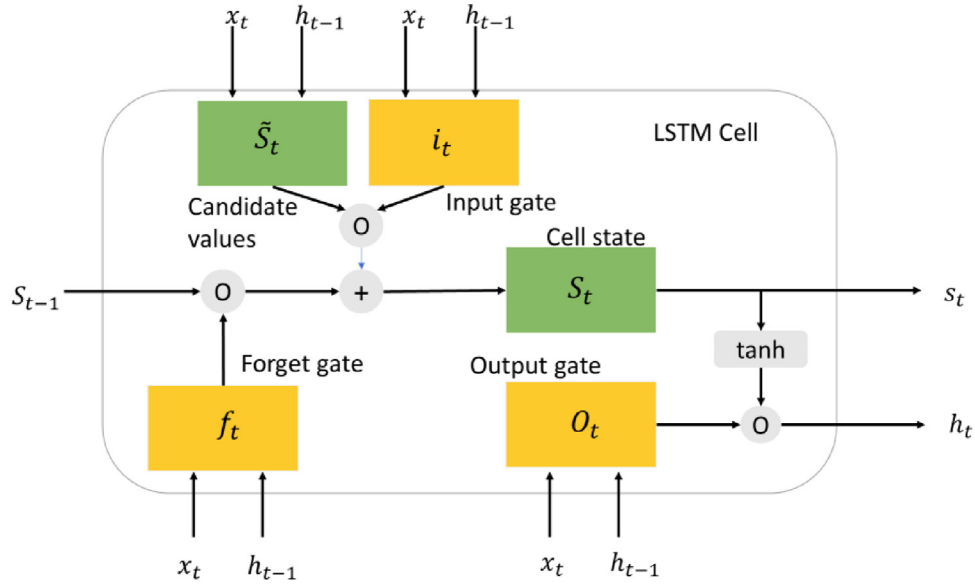


Fig. 2. Structure of LSTM memory cell.

$$(2) \text{ MAE} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|$$

$$(3) \text{ MAPE} = \frac{1}{T} \sum_{t=1}^T \left| 1 - \frac{\hat{y}_t}{y_t} \right|,$$

$$(4) R^2 = \frac{\sum_{t=1}^T (y_t - \bar{y})^2}{\sum_{t=1}^T (\hat{y}_t - \bar{y})^2}$$

where y_t and $\hat{y}_t$ are the actual and predicted prices of Bitcoin, respectively, at time t with total observations T , and $\bar{y}$ indicates the average of all observations. RMSE squares the difference between predicted and actual values, and is therefore sensitive to outliers. However, MAE is more robust to outliers because it only measures the difference between predicted and actual values. MAPE calculates the relative ratio of the error to the actual value as a percentage from 0~100, making intuitive interpretation easy. Finally, R^2 indicates the proportion of variance for actual values that can be explained by the predicted values.

(2) Classification problem:

$$(1) \text{ Sensitivity} = \frac{TP}{TP+FN},$$

$$(2) \text{ Specificity} = \frac{TN}{TN+FP},$$

$$(3) \text{ Accuracy} = \frac{TP+TN}{TP+FN+TN+FP},$$

$$(4) \text{ Balance accuracy} = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$

where TP indicates an event for a turbulent state of the Bitcoin market correctly classified as a turbulent state, FN is a turbulent regime misclassified as a non-turbulent state event, and FP is a tranquil state event misclassified as a turbulent state event. Finally, TN implies a tranquil state event correctly classified as a tranquil state. Sensitivity is the ratio of instances correctly classified as turbulent among actual instances so labeled, while specificity is the ratio of instances correctly classified as tranquil. Accuracy indicates the proportion of all instances that are correctly classified regardless of a certain class. Since sensitivity and specificity are inversely proportional, balance accuracy measures the average performance of the two.

(5) AUC: the ROC curve is a two-dimensional graph in which sensitivity is plotted on the y-axis and 1 minus specificity is plotted on the x-axis. The ROC curve explains the relative trade-off between gain and costs with changes in probability thresholds. When the threshold classifying the event of a turbulent state increases, the sensitivity increases and the specificity decreases,

whereas when the threshold decreases, the opposite occurs. The area under the ROC curve is AUC, and the larger the AUC, the better the performance.

3.3.1. The basic LSTM model

The LSTM network suggested by Hochreiter and Schmidhuber (1997) [91] is used in this study as a representative recurrent neural network for sequenced learning tasks. The LSTM uses memory cells to maintain long sequences of inputs and predict the next step in the sequence of observations with respect to those in previous steps. The structure of the LSTM memory cell is described in Fig. 2.

At time t , a memory cell has the following three gates to retain its cell state: (1) a forget gate f_t , (2) an input gate i_t , and (3) an output gate o_t . The execute steps of the LSTM cell are summarized as follows.

In the first step, given input x_t at time t , activation values f_t in the forget gate determine which information is removed from the previous cell state h_{t-1} at previous time $t-1$, computed as:

$$f_t = \text{sigmoid}(W_{f,x}x_t + W_{f,h}h_{t-1} + b_f). \quad (5)$$

The second step is to determine how much to update the current time-series information in the new cell state. The activation values i_t of the input gate determine what candidate values $\tilde{S}_t$ in the given input are to be updated in the current cell state S_t , computed as follows:

$$\tilde{S}_t = \tanh(W_{\tilde{S},x}x_t + W_{\tilde{S},h}h_{t-1} + b_{\tilde{S}}) \text{ and} \quad (6)$$

$$i_t = \text{sigmoid}(W_{i,x}x_t + W_{i,h}h_{t-1} + b_i). \quad (7)$$

In the third step, the cell states S_t are obtained by a combination of the previous cell state S_{t-1} and the current candidate values $\tilde{S}_t$, computed as:

$$S_t = f_t \times S_{t-1} + i_t \times \tilde{S}_t, \quad (8)$$

where multiplication at the front determines how much previous information is forgotten, and multiplication at the back indicates how much current information is maintained.

Algorithm 1. Proposed AdaBoost-LSTM algorithm

D: Training data set $\{(x_i, y_i)\}_{i=1}^k$ with size k , where y_i is the target variable for Bitcoin price movement m_i and
Input: turbulent regime h_i of the Bitcoin market. h_i is a positive and negative sample as $[1, -1]$.
N: Number of LSTM learners to be included in the ensemble.

Aims: 1) Strong regressor for Bitcoin price movement and 2) Strong classifier for turbulent volatility regime

```

1: Step 1. Generating normalized training data set.
2:   for  $i=1:k$  do
3:      $Normalized\ indicator_i = \text{Min-Max}(x_i)$ 
4:      $Q = \text{Stack}(Normalized\ indicator_i)$ 
5:   end for
6:    $D = \text{Merge}(Q, Y)$ 
7: Step 2. Initialize weights  $w$  of samples  $x_i$  in training set  $D$ .
10:   $D_1 = (w_1^1, w_1^2, \dots, w_1^k)$ ,  $w_1^i = \frac{1}{k}$  where  $i=1, 2, \dots, k$ .
14: Step 3. Training a weak model,  $H_n(x)$  with inputs of the training set  $D$  and weights set  $D_n$ .
16:   for  $n=1:N$  do
17:     Step 4. Train using LSTM framework as a base learner,  $H_n(x) = \text{LSTM}_n(D, D_n)$ .
18:     Step 5. Calculate the weighted estimation error of samples  $x_i$  combined with respect to the weights  $w_n^i$ .
        /* Eq(11-12) and Eq(15) are formulae for regressor and classifier, respectively.
19:     Step 6. Obtain model confidence  $\alpha_n$  and update the weights set of samples  $x_i$  as  $D_{n+1}$ .
        /* Eq(13-14) and Eq(16-17) are formulae for regressor and classifier, respectively.
20:   end for

```

Output: Strong ensemble models, $H(x) = \sum_{n=1}^N \text{confidence } \alpha_n * H_n(x)$.
 /* Eq(18) and Eq(19) are formulae for regressor and classifier, respectively.

In the last step, the output h_t is dependent on the activation values O_t of the output gate, which is computed as:

$$O_t = \text{sigmoid}(W_{O,x}x_t + W_{O,h}h_{t-1} + b_O) \text{ and} \quad (9)$$

$$h_t = O_t \times \tanh(S_t). \quad (10)$$

In Eqs. (5)–(10), $W_{f,x}$, $W_{f,h}$, $W_{\tilde{S},x}$, $W_{\tilde{S},h}$, $W_{i,x}$, $W_{i,h}$, $W_{O,x}$, and $W_{O,h}$ indicate weight matrices, b_f , $b_{\tilde{S}}$, b_i , and b_O are bias vectors, and the sigmoid function is used as the activation function. During learning, the weights W and bias terms b are optimized to minimize the error of the loss function.

3.3.2. Proposed fusing prediction approach

(1)) AdaBoost-LSTM ensemble model

In this section, we devise a way to prevent overfitting while extracting elaborately high-dimensional patterns in the presence of many variables, solving the problem using the ensemble framework. In particular, we adopt the AdaBoost ensemble framework proposed by Freund and Schapire [31]. Adaptive sequential learning using AdaBoost not only minimizes overfitting, but also continuously improves the prediction errors of the models to be combined. We utilize an AdaBoost-LSTM ensemble model including multiple LSTMs as learners in the AdaBoost framework. Based on the ensemble framework, we build a two-state prediction model with a Bitcoin price regressor and turbulent regime classifier. In particular, the turbulent regime classifier is an information fusion approach that uses Markov modeling of turbulence information.

The architecture of the proposed AdaBoost-LSTM is shown at the bottom of Fig. 1 above. Fig. 1 illustrates the generation of multiple sequential LSTMs based on repeatedly reweighted training data. The final output is the sum of the predicted outputs of each model combined with the weight (= each model's credibility). The whole process is expressed in Algorithm 1.

Let normalized training samples D with size k and the AdaBoost algorithm generate N sub-models based on an LSTM learner. Let it train each model H_n based on a combination of the original set D and weights D_n . The weight is combined with the predicted outputs from each trained model, and is updated

during the boosting process according to the prediction errors. Specifically, for a predictor of Bitcoin price y_i , the normalized estimation error ϵ_n^i for sample x_i is written as:

$$\epsilon_n^i = \frac{|y_i - H_n(x_i)|}{\max_{i=1}^k |y_i - H_n(x_i)|}, \quad (11)$$

and the weighted estimation error ϵ_n is computed as:

$$\epsilon_n = \sum_{i=1}^m w_n^i \epsilon_n^i. \quad (12)$$

Then, the weights w_{n+1}^i of sample x_i is updated for the next model H_{n+1} according to model confidence α_n , which is calculated as:

$$\alpha_n = \eta \times \frac{\epsilon_n}{1 - \epsilon_n} \text{ and} \quad (13)$$

$$w_{n+1}^i = \frac{w_n^i \alpha_n^{1-\epsilon_n^i}}{Z_n}, \quad (14)$$

where η is the learning rate = 1, and Z_n indicates a normalized constant which enables $\sum_{i=1}^k w_{n+1}^i = 1$.

Given sample x_i and its output $y_i \in \{0, 1\}$, a classifier of the high-volatility regime H_n follows the estimation error ϵ_n as:

$$\epsilon_n = \frac{\sum_{i=1}^k w_n^i \Pr[y_i \neq H_n(x_i)]}{\sum_{i=1}^k w_n^i}, \quad (15)$$

where w_n^i and $\Pr[y_i \neq H_n(x_i)]$ are the weight and the misclassification probability of x_i , respectively. The weights are updated similarly to Eqs. (13)–(14), which are computed as:

$$\alpha_n = \eta \times \frac{1}{2} \ln \left(\frac{1 - \epsilon_n}{\epsilon_n} \right) \text{ and} \quad (16)$$

$$w_{n+1}^i = \frac{w_n^i}{Z_n} \times \begin{cases} \exp(-\alpha_n), & \text{if } h_i = H_n(x_i) \\ \exp(\alpha_n), & \text{if } h_i \neq H_n(x_i). \end{cases} \quad (17)$$

Intuitively, we see that the larger the error ϵ_n of the previous model, the lower the model confidence α_n . Therefore, high weight is assigned to improve the prediction error in the next model.

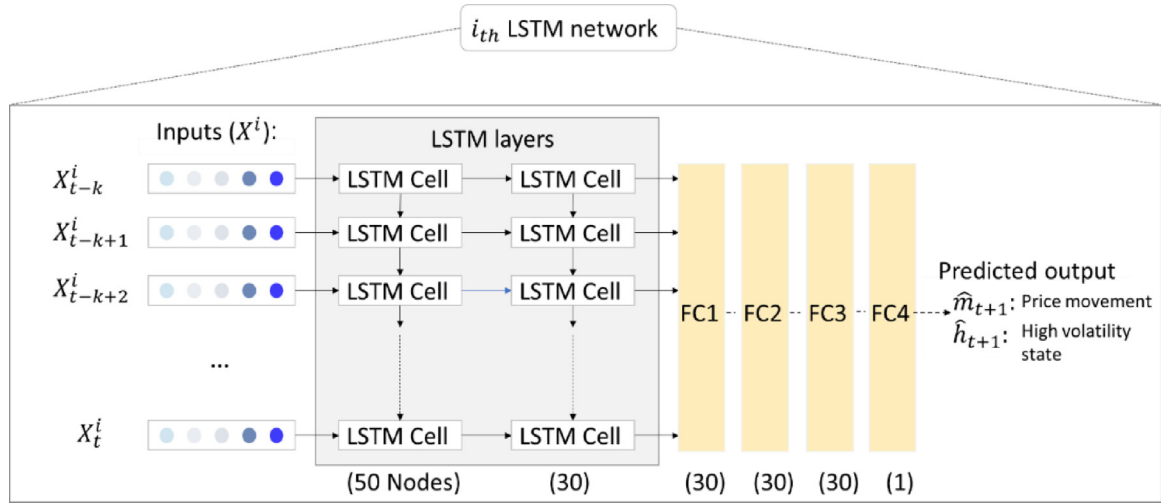


Fig. 3. The architecture of our LSTM networks.

Thus, our AdaBoost-LSTM methodology reduces estimation errors by paying attention to training samples that are difficult to predict accurately.

The final model is constructed by summing the sub-model trained with reweighted samples in each iteration as follows:

Bitcoin price regressor:

$$H_{price}(x) = \sum_{n=1}^N \left[\frac{1}{2} \ln \left(\frac{1 - \epsilon_n}{\epsilon_n} \right) \times H_n(x) \right] \text{ and} \quad (18)$$

Bitcoin turbulent regime classifier:

$$H_{regime}(x) = \sin \left(\sum_{n=1}^N \left[\frac{1}{2} \ln \left(\frac{1 - \epsilon_n}{\epsilon_n} \right) \times H_n(x) \right] \right), \quad (19)$$

where $\ln \left(\frac{1 - \epsilon_n}{\epsilon_n} \right)$ indicates model confidence, and the higher the confidence of the sub-model, the more the predicted output is reflected in the final output, and the lower the confidence, the less it is reflected.

For the Bitcoin turbulent regime classifier, the final model is a binary function that outputs a value of 1 if the result exceeds 0.5 and -1 otherwise, expressed as:

$$H_{regime}(x) = \begin{cases} 1, & \text{if } H_{regime}(x) > 0.5 \\ -1, & \text{otherwise.} \end{cases}$$

Fig. 3 shows the architecture of the LSTM network used as a base learner in the AdaBoost framework. We use two LSTM layers with 50 and 30 neurons, respectively. Next, three fully connected layers with 30 neurons, and 1 fully connected layer with 1 neuron for output are attached. For the final output, the linear activation function (Relu) is used for Bitcoin price prediction and the sigmoid function is used for the direction and high-volatility regime in the Bitcoin market.

In addition, we adjust the time steps according to the forecast horizon: 5 days timesteps for the 1- and 5-day forecast horizon models, and 7 days timesteps for 3-day forecast horizons model. The given dropout and learning rates are 0.2 and 0.001, respectively, and an Adam optimizer is used. All software is implemented through open-source Keras³. (See details such as

hyperparameter tuning of our LSTM model and other benchmark models in Appendix A.)

Based on the AdaBoost-LSTM framework, we construct two heterogeneous ensemble models; (1) Bitcoin price prediction model and (2) turbulence regime prediction in the Bitcoin market. The key of our work lies in how to fuse heterogeneous ensemble models built for different purposes to further improve the prediction accuracy of the Bitcoin market.

(2) Intelligent Bitcoin trading system

Finally, we build a Bitcoin trading system that can maximize profitability based on the potential inherent features of the Bitcoin market obtained by fusing partial predictions such as direction and turbulence regime. We design a differentiated trading strategy by subdividing the forecast horizon into short-term (1-day) and medium-term (3-day and 5-day) forecast horizons. Details of the trading algorithm are specified in Algorithm 2.

In Step 1 of our Bitcoin direction model, output d_t at the current time t is classified as 1 if the Bitcoin price is predicted to be arising in an up direction with a high probability at time $t + f$, after the forecast horizon f from time t . Conversely, if $d_t = -1$, it indicates a downward direction. That is, d_t is predicted output of whether the Bitcoin return is positive or negative.

In Step 2, There are three transaction states: Buy, Sell, and Hold current position, and the transaction state at time t changes when the previous direction $d_{t-1} \neq$ current direction d_t . For example, if $\{d_t = 1, d_{t-1} = -1\}$, the state of the trading transaction changes 'Buy,' and the opposite is the 'Sell' signal. However, if the previous and current directions are the same, there is no change in the trading transaction. For example, if a "buy" signal is given the previous day and a continuous rise is predicted even today, the system gives a "hold" signal. It is the investor's decision whether or not to buy more Bitcoin.

In Step 3, when the turbulent period in the Bitcoin market is prolonged more than L days in a row (we call this the market turbulence period), the transaction is unconditionally converted to a "Sell" signal of -1 during the turbulent regime period j that continues from the next day with consideration of uncertainty.

In Step 4, we demonstrate the superiority of the proposed system by comparing the cumulative returns of our trading system and traditionally widely used auxiliary indicators such as MACD, ROC, and the stochastic oscillator.

³ Francois Chollet. Keras. GitHub repository: <https://github.com/fchollet/keras>, 2015.

Algorithm 2. Proposed intelligent trading system

Input: D^f : Predicted outcomes $\{(d_t, h_t)\}_{t=1}^T$ with time $t = 1, 2, \dots, T$, based on forecast horizon $f = \{1, 3, 5\}$, where d_t and $h_t \in [-1, 1]$ indicate Bitcoin price direction and turbulent regime, respectively.

- 1: **Step 1.** Select the forecast horizon $f = \{1, 3, 5\}$ days for short- to medium-term trading strategy.
- 2: **Step 2.** Follow trading transaction based on the predicted direction d_t at time t .
- 3: Transaction state(t) = $\begin{cases} \text{Buy, if } d_t = 1 \text{ and } d_{t-1} = -1 \\ \text{Sell, if } d_t = -1 \text{ and } d_{t-1} = 1 \\ \text{Hold current portfolio if } d_t \text{ equal to } d_{t-1}. \end{cases}$
- 4: **Step 3.** Prohibit buying when prolonged period of turbulent regime continues L days in a row.
- 5: Direction $\{d_t\}_{t+1}^j = -1$, if $\sum_{t=t-L}^t h_t \geq L$, where the period j in which h_t is 1 continuously from time t
- 6: **Step 4.** Obtain cumulative profit of our proposed trading algorithm.
- 7: Cum(t) = $\ln(\text{close price}(t+f)) - \ln(\text{close price}(t))$ during Steps 2-4.
- 8: **Step 5.** Obtain hit ratio of our proposed trading algorithm from the past N -day to the time t
- 9: Hit ratio(t) = $\frac{1}{N} \sum_{t=t-N}^t p_t$ ($N = 7, 14, 30$)
- 10: $p_t = \begin{cases} 1, & d_t \times (\text{actual price}_{t+f} - \text{actual price}_t) > 0 \\ 0, & \text{otherwise} \end{cases}$

Output: Prediction-based Bitcoin trading system

Table 4

Summary of cutoff points for turbulent regimes in the full sample and test set.

	(a) Probability of turbulent regime				(b) Regime in Bitcoin market			
	Mean	Median	SD	Skew	Turbulent regime		Tranquil regime	
					Period (days)	Average log return (%)	Period (days)	Average log return (%)
Cutoff _{Full-sample}	0.53	0.59	0.41	-0.12	1156	0.23	1004	0.19
Cutoff _{Test}	0.75	0.52	0.35	-0.45	247	-0.25	148	0.14

In Step 5, we calculate the hit ratio to measure how well our trading signal matches the actual price direction of Bitcoin. The hit ratio is obtained by multiplying the predicted output with the actual price direction. If it is greater than 0, it is classified as a hit event; otherwise, miss. The hit ratio shows the cumulative ratio of the number of hits or misses for the previous 7 days, 14 days, and 30 days. Therefore, the investors can check the hit ratio in real time and make decisions such as using a forecast horizon with many hit ratios.

4. Empirical analysis

4.1. Modeling stage I

4.1.1. Turbulent regime thresholding

We classify the time period of the turbulent regime in the Bitcoin market by setting the cutoff point of the filtering probability based on historical log returns. For the two-state problem, previous studies [27] set the cutoff point as the minimum value between two peaks of a frequency density histogram. Fig. 4 shows the frequency density histogram of the probability of high-volatility regimes in the full sample from June 1, 2016 to April 30, 2022. In Fig. 4, the cutoff point of the entire sample is skewed to the right, with a median value of 0.59 greater than the minimum value of 0.49 and a mean value of 0.53.

Table 4 provides a summary of the cutoff point. In Table 4(a), negative skewness indicates that the cutoff is higher than the minimum and more likely to take values around the mean and median. Therefore, in our study, the mean value of 0.53 better fits the cutoff point than the minimum value of the density histogram. In addition, if the probability of a high-volatility regime exceeds the mean value, the dummy variable of the turbulent

regime is assigned a value of 1, otherwise 0. Table 4(b) shows a summary of regimes in the Bitcoin market classified according to a cutoff point of 0.53.

Finally, Fig. 5 shows the periods of turbulence in the Bitcoin market. Fig. 5(b) shows the probability of a turbulent regime, and Fig. 5(c) shows periods of turbulence (shaded area).

Table 5 provides a summary of statistics for periods of turbulent regime lasting more than 7 consecutive days. The proportion of turbulent regime in the training set is 46.4% of the total, which is smaller than the tranquil period at 53.7%, but not to the extent that it causes data imbalance problems. (Additional details and description of market turbulence in the Bitcoin market are available in Appendix B.)

In addition, we verify statistically significant differences in the values for our dummy variables between periods of turbulence and tranquility via t -test, as shown in Table 6(a).

4.1.2. Impact of turbulence on the bitcoin market

In this section, we analyze the impact of turbulence in the Bitcoin market. As shown in Table 4(b), the periods of turbulence are 1156 and 247 days for the full sample and test set, respectively. In addition, the average daily return during periods of turbulence is a positive value of 0.23% for the full sample, which is higher than that of 0.19% during the tranquil period. Thus, a continuation of the period of turbulence generally leads to high returns. However, during the period of turbulence in the test sample, a negative return (-0.25%) is observed, in stark contrast to a positive return (0.14%) during the tranquil period.

We now analyze how a turbulent regime affects the Bitcoin market with a time lag. Table 6(b) shows the influence of the probability of a turbulent regime on Bitcoin returns using OLS regression. In Table 6(b), the positive coefficient value of the

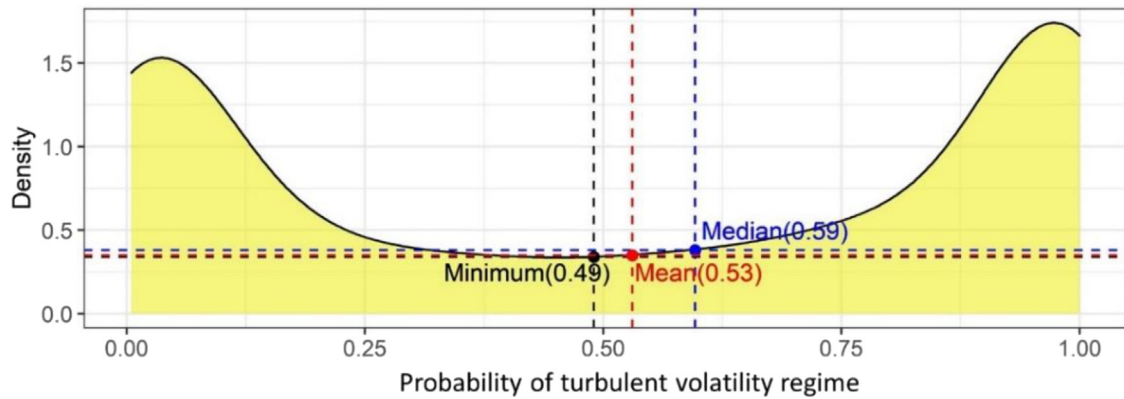
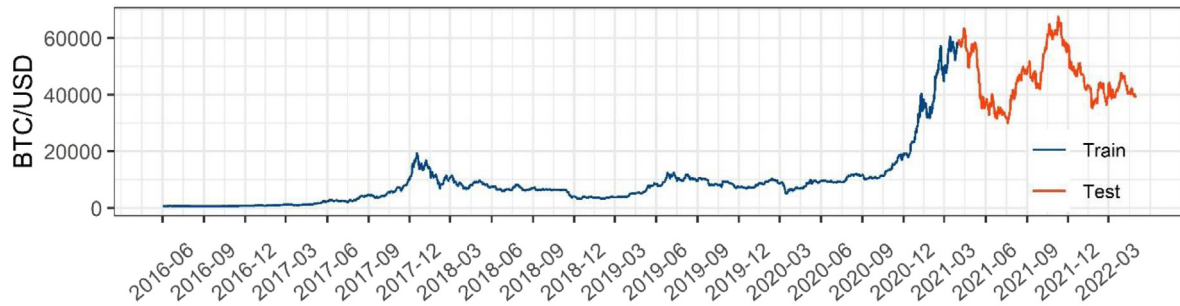
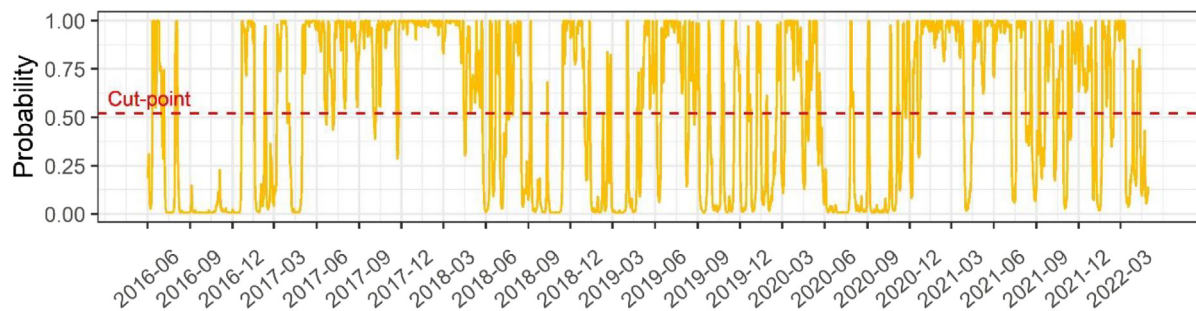


Fig. 4. Density for high-regime probability obtained using the Markov regime-switching model.

(a) Bitcoin prices



(b) Probability of turbulent regime



(c) Periods of turbulence

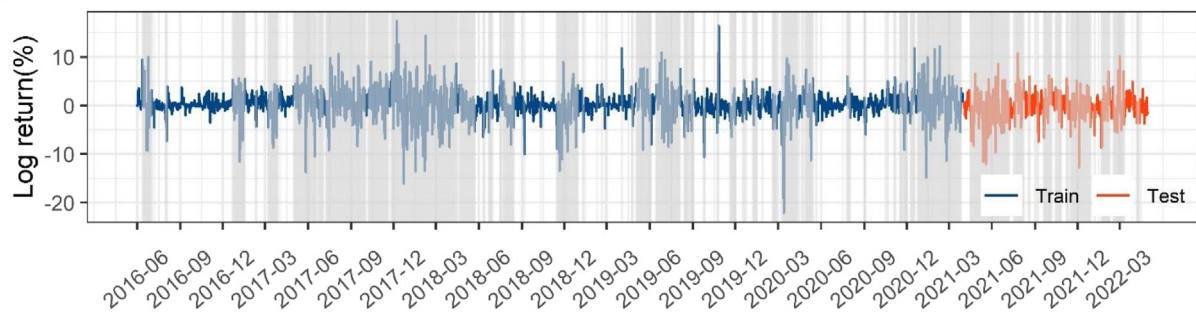


Fig. 5. Periods of turbulence in Bitcoin market (gray shaded area).

Table 5

Summary of statistics for periods of turbulent regime lasting more than 7 consecutive days.

	Period	N	Turbulent regime	Tranquil regime	Duration			
					Min	Max	Median	Average
Train	2016.6.1 – 2021.10.5	1588	736(46.4%)	852 (53.7%)	7	141	23	32.0
Valid	2021.10.6 – 2022.3.31	177	119 (67.2%)	58 (32.8%)	8	94	15	31.75
Test	2021.4.1 – 2022.4.30	395	228 (57.7%)	167 (42.3%)	7	84	20	28.5

Table 6Effect of turbulent regime periods in the Bitcoin market: (a) Results of *t*-test to account for unequal variances and mean differences in macro-level and blockchain variables between tranquil and turbulent regimes, and (b) OLS regression on the effect of the probability of turbulent regime on Bitcoin returns according to lagged time.

(a) Results of the <i>t</i> -test				(b) Results of OLS regression		
Variables	Periods		<i>t</i> -stat	Lagged time	Logarithmic returns	
	Tranquil regime	Turbulent regime			Coefficient	<i>t</i> -stat
10-year bonds	1.888	1.98	−2.96***	<i>k</i> = 1	0.0412	2.212**
2-year bonds	1.3	1.268	0.84	<i>k</i> = 2	0.0843	1.883**
Gold	1546.84	1510.11	3.01***	<i>k</i> = 3	0.1108	4.035***
S&P 500	3029.75	3157.4	−3.98***	<i>k</i> = 4	0.1024	1.917**
Nasdaq	8794.26	9341.43	−3.98***	<i>k</i> = 5	0.0658	1.012
VIX	17.926	18.579	−1.86*	<i>k</i> = 6	0.1282	6.915***
Dollar index	96.487	94.723	13.63***	<i>k</i> = 7	0.1678	3.048**
WTI	56.439	58.295	−2.79***	<i>k</i> = 8	0.1209	1.355*
Tweets	41888.8	68500.1	−13.48***	<i>k</i> = 9	−0.0004	−1.322
Price	12301	18230.3	−8.07***	<i>k</i> = 10	−0.05	−2.312**
Log return	0.189	0.228	8.45***	<i>k</i> = 11	−0.0607	−4.378***
Block size	1.018	1.07	−6.02***	<i>k</i> = 12	−0.0368	−2.229**
Market capital	226955	335139.2	−7.78***	<i>k</i> = 13	−0.2113	−1.07
Number of transactions	0.278	0.276	0.72	<i>k</i> = 14	−0.3247	−7.154***
Trade volume	341.55	1009.64	−20.97***	<i>k</i> = 15	−0.1761	−1.254
Hash rate	90.28	76.755	4.99***			

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

turbulent regime variable increases as time passes within 8 days of the lagged time *k*, which indicates that Bitcoin returns increase significantly during the initial turbulent regime period. However, surprisingly, as lagged time *k* increases past 9 days, the negative coefficient of the turbulent regime variable increases, which indicates that Bitcoin returns are likely to plummet in the mid to long term during prolonged periods of turbulence.

We presume that these findings could be related to the bubble phenomenon. A bubble is characterized by an increase in speculative funds and positive feedback resulting from herding behavior, after which prices increase exponentially and then plunge suddenly [92]. This is true in the Bitcoin market. Many researchers [93,94] have detected several bubble periods in the medium to long term, at the end of which catastrophic price declines were found.

In conclusion, we hypothesize that if we predict a turbulent regime using the proposed methodology and subsequently refrain from buying Bitcoin during prolonged periods of turbulence, higher returns can be obtained.

4.2. Modeling stage II

4.2.1. Optimal design of AdaBoost-LSTM model

Determining the optimal number of LSTMs to be included in the AdaBoost framework is important. We find the optimal numbers of LSTMs to be $N = 2$ to 50. In addition, the training data for each LSTM model is sequentially chosen with replacement from 90% of the total training set, and the remaining 10% as the validation set is used to determine the hyperparameters.

We built two types of prediction models for a specific purpose: (1) an AdaBoost-LSTM regressor to predict Bitcoin prices, and (2) an AdaBoost-LSTM classifier to predict turbulence in the Bitcoin market. We set the forecast horizons of the model as 1-day, 3-days, and 5-days. Bitcoin's volatility is much higher than other financial assets such as the stock index and exchange rate, so

predicting beyond 5-days can reduce accuracy. In addition, many previous studies have argued that Bitcoin is highly useful as an asset that can hedge short-term risks within 3–5 days [95,96].

Fig. 6 shows the performance of the two prediction models according to changes in the number of ensemble combinations. This figure also represents differences according to the forecast horizon. Overall, as the number of ensemble combinations increases, performance improves compared to the LSTM model alone ($N = 0$). Specifically, for the 1-day forecast model, the RMSE error in Fig. 6(a) is reduced by up to −40.2% compared to the LSTM model alone, and the AUC of classification performance in Fig. 6(b) is up to 13.3% higher than that of the LSTM alone.

However, performance improvement does not change significantly after the number of combinations *N* exceeds 30; thus, in this study, we recommend that the optimal number of LSTM combinations is between 30 and 50. (See Appendix C for a detailed comparison using various measurement indicators.)

4.2.2. Comparison with baseline models

Next, we compare the predictive performance between our proposed model and baseline models such as the least absolute shrinkage and selection operator (LASSO) regression model, the AdaBoost base model based on a decision tree, and the LSTM alone. The LASSO regression model, proposed by Tibshirani (1996) [97], addresses the multicollinearity problem by reducing the coefficients of variables with low influence using regularization. It is used as the best linear model for comparison with ML models. XGBoost and GRU models were used as benchmarking models for AdaBoost and LSTM models, respectively.

Table 7(a) indicates that XGBoost has a slightly smaller prediction error than AdaBoost, but the difference is not significant. GRU model has generally higher prediction error than LSTM. However, our AdaBoost-LSTM regressor has a lower prediction error than all other baseline models and has excellent explanatory power, as indicated by the high R^2 value.

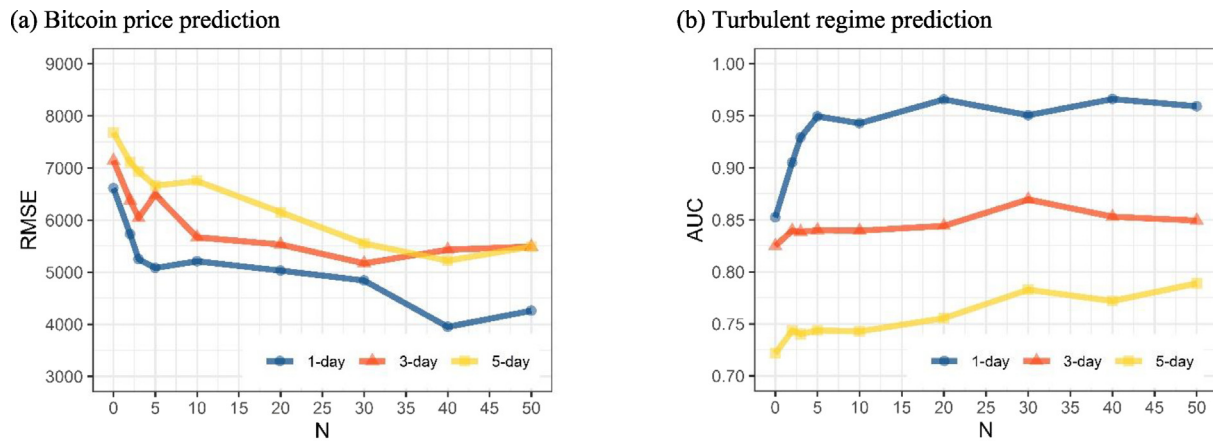


Fig. 6. Predictive performance of ensemble models trained with different numbers of LSTM learners: (a) Bitcoin price regressor and (b) turbulent regime classifiers.

Table 7

Comparison of prediction models for Bitcoin prices according to different forecast horizons. Diebold–Mariano test [98] was used to account for statistical differences between model types.

Forecast horizon	Model	(a) Measurement				(b) Diebold–Mariano test
		MAE	RMSE	MAPE	R^2	
1 day	LASSO	7180	7570	15.232	0.267	24.432***
	AdaBoost _{Base}	3540	4270	8.595	0.766	1.896*
	XGBoost	3410	4830	6.558	0.701	1.453
	LSTM _{Base}	5480	6610	11.285	0.420	13.649***
	GRU	6890	8140	13.98	0.542	13.03***
	AdaBoost–LSTM	3260	3950	7.010	0.797	–
3 days	LASSO	8440	9080	17.817	0.043	33.826***
	AdaBoost _{Base}	5370	6010	12.654	0.535	4.113***
	XGBoost	5110	5900	11.92	0.551	4.01***
	LSTM _{Base}	5740	7140	11.563	0.332	12.437***
	GRU	5620	7000	11.280	0.551	8.61***
	AdaBoost–LSTM	4150	5170	8.754	0.650	–
5 days	LASSO	9820	10600	20.681	0.012	19.942***
	AdaBoost _{Base}	7470	8390	17.976	0.090	13.025***
	XGBoost	6880	7930	16.663	0.188	9.832***
	LSTM _{Base}	6110	7680	12.863	0.199	8.404***
	GRU	6990	8870	14.988	0.130	11.65***
	AdaBoost–LSTM	4320	5220	9.393	0.645	–

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively; p -values below 0.05 indicate that the null hypothesis of equal classifier performance is rejected at the 0.05% level.

In particular, the RMSE error of AdaBoost–LSTM decreases by –53.8% on average for all forecast horizons compared to the LASSO linear model. In addition, as the forecast horizon increases, the prediction error increases, whereas in our model, there is no significant difference in performance between short- (1-day) and medium-term (3 or 5-day) forecast horizons compared to the baseline model. Since our proposed model utilizes an ensemble method that combines multiple models while continuously accounting for prediction error, it minimizes the loss of predictive power.

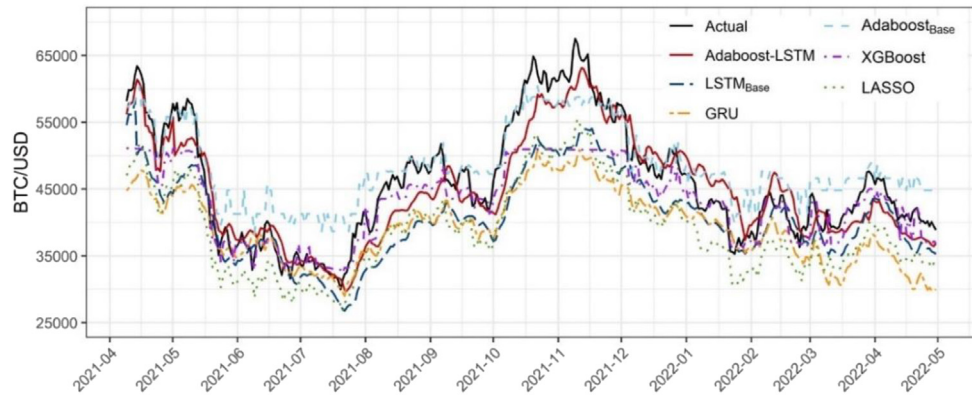
Furthermore, we verify the significance of differences in performance between the proposed model and the baseline model using a Diebold–Mariano (DM) test [98]. The basic idea of this test is to compare the expected loss differential between two forecast errors. The DM test is based on a null hypothesis that there is no difference between two prediction errors. If the p -value of the DM statistic is less than 0.05, the null hypothesis is rejected, indicating that the two predictive models are statistically different. This test has been widely used in ML studies because it has the advantage of considering sampling variability and mean loss [99]. The DM test results in Table 7(b) show that there is a statistically significant difference in forecast error (RMSE) between our model and the baseline models.

Fig. 7 plots the time-series forecasts and actual values of the models during the test set period. Fig. 7 shows that the predicted values of our AdaBoost–LSTM model (red line) most closely follow the actual values (black line) among all forecasts. In addition, for the 5-day forecast horizon in Fig. 7(c), the baseline models deviate greatly from the actual values, but our model still predicts the trends closest to the actual values.

Next, we compare the prediction accuracy of classifiers for turbulent regimes in the Bitcoin market. The results in Table 8(a) demonstrate that the prediction accuracy of the AdaBoost–LSTM proposed model is higher than that of other baseline models.

In addition, we verify statistical differences in classification performance between our model and the baseline models using the test developed by DeLong et al. (1988) [100]. The DeLong test is a non-parametric analysis of two or more AUC values. Parametric tests such as the t -test and ANOVA are used for verification of statistical differences between several groups, but they are not suitable for use in ML studies because the assumptions of parametric tests are likely to be violated in comparisons of ML classifiers [101]. A p -value of the DeLong statistic less than 0.05 indicates that the null hypothesis is rejected and that there is no difference in AUC between the two classifiers. The results

(a) 1-day forecast horizon



(b) 3-day forecast horizon



(c) 5-day forecast horizon



Fig. 7. Time-series forecasts of each model for Bitcoin prices according to different forecast horizons: (a) 1-day, (b) 3-day, and (c) 5-day forecast horizons.

of the DeLong test in Table 8(b) reveal that the differences in AUC performance between our model and the reference model are mostly statistically significant.

We now plot the ROC curve, as shown in Fig. 8, to observe the AUC performance of each classifier in detail. The ROC curve represents the relationship between sensitivity (= share of correctly classified turbulent regimes) and 1 minus specificity (= false warnings) according to changes in the threshold of probability in the process of classifying a given instance as a turbulent regime or not. Since sensitivity and specificity are inversely proportional to each other, when sensitivity is increased by raising the threshold, more false warnings occur. Therefore, maintaining both indicators at a high level is difficult.

In Fig. 8, our proposed AdaBoost-LSTM model has significantly wider AUC values than those of baseline models. In particular, the false alarm rate is lower than other models, although a high level of sensitivity is maintained. For example, for the 1-day forecast horizon, our AdaBoost-LSTM shows a low false warning rate of 0.05 even at a high sensitivity level of 0.8, whereas the false warning rate more than quadruples from 0.2 to 0.3 when the baseline model maintains the same level of sensitivity. Similar performance characteristics are observed for the (b) 3-day and (c) 5-day forecast horizons. Therefore, we conclude that our ensemble approach allows us to construct a stable model with a low false positive rate while maintaining high sensitivity. Finally, we compare the robustness of our AdaBoost-LSTM model and the

Table 8

Comparison of classification models for turbulent regimes according to different forecast horizons. DeLong test [100] was used to account for statistical differences between model types.

Forecast horizon	Model	(a) Measurement				(b) DeLong test
		Accuracy	Sensitivity	Specificity	AUC	
1 day	LASSO	0.765	0.814	0.681	0.794	6.443***
	AdaBoost _{Base}	0.835	0.891	0.742	0.875	4.114***
	XGBoost	0.835	0.907	0.712	0.890	1.971*
	LSTM _{Base}	0.789	0.776	0.816	0.852	4.893***
	GRU	0.791	0.891	0.617	0.870	3.256***
	AdaBoost-LSTM	0.930	0.996	0.816	0.966	–
3 days	LASSO	0.728	0.785	0.629	0.750	3.703***
	AdaBoost _{Base}	0.702	0.652	0.786	0.817	1.826*
	XGBoost	0.704	0.676	0.752	0.822	1.875*
	LSTM _{Base}	0.778	0.909	0.537	0.825	2.246**
	GRU	0.746	0.866	0.532	0.801	3.372***
	AdaBoost-LSTM	0.816	0.901	0.647	0.870	–
5 days	LASSO	0.665	0.737	0.539	0.667	3.254***
	AdaBoost _{Base}	0.697	0.741	0.622	0.741	1.387*
	XGBoost	0.625	0.603	0.662	0.737	1.533
	LSTM _{Base}	0.712	0.789	0.571	0.722	1.759*
	GRU	0.628	0.607	0.664	0.710	2.333**
	AdaBoost-LSTM	0.759	0.875	0.554	0.789	–

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively; p -values below 0.05 imply that the null hypothesis of equal classifier performance is rejected at the 0.05% level.

Table 9

Learning time by each prediction methodology. All modes are operated under the following conditions: Intel Core i7 (CPU 4-core) 2.6 GHz processor, 16 GB memory, and Iris Plus Graphics.

Model	Time (second)	Model	Number of ensembles	Time (second)
LASSO	0.079	AdaBoost-LSTM	2	32.145
AdaBoost _{base}	0.303		3	52.048
LSTM _{base}	20.221		5	79.235
GRU	12.336		10	196.871
XGB	0.103		20	407.095
			30	730.365
			40	1199.439
			50	1446.169

LSTM model alone by measuring differences between the training and test set errors. Fig. 9 shows that the difference between training and test errors in our AdaBoost-LSTM model is much smaller than that of the LSTM model alone. Thus, we estimate that our ensemble framework has significantly less error due to overfitting.

Finally, we measured the learning time of each model, as shown in Table 9. For a single model, the LSTM and GRU models took the longest to learn due to their large number of parameters. With AdaBoost-LSTM, we observed that the number of ensembled LSTMs and learning time had an exponential relationship, which is a disadvantage of this approach. AdaBoost-LSTM learning is associated with long learning times.

4.2.3. Implementation of an intelligent trading system

In this section, we describe an intelligent trading system that classifies the direction of the Bitcoin market based on knowledge of the turbulent regime, verifying through trading simulations that our proposed model can be applied to an actual market. The classifier is learned by the AdaBoost-LSTM framework, which is an ensemble of 40 LSTMs. Our trading system identifies movement in the Bitcoin market in an upward direction (1) if the probability exceeds 0.5 and movement in a downward direction (0) otherwise.

We compare our trading system with popular trading indicators such as MACD, ROC, and stochastic oscillators in terms of direction prediction accuracy and profitability. We also compare it to the traditional long-only strategy of holding since the initial purchase. (Appendix D shows a description of parameters and “buy” signal conditions according to other trading indicators.)

Table 10 shows the classification accuracy of our proposed model and other trading indicators for the direction of movement of Bitcoin prices during the period of the test set. While the accuracy of the long-only holding strategy was the lowest, our AdaBoost-LSTM model had the highest accuracy; in particular, the classification accuracy for the medium-term (3- or 5-day forecast horizons) was better than the short-term (1-day forecast horizon).

Based on our prediction model for the turbulent state of the Bitcoin market, we establish a more advanced trading strategy that defines a long-lasting turbulent regime as a market turbulence period and prohibits buying during a period of market turbulence. We simulate on a weekly basis from 0 to 28 days to find periods of market turbulence that can result in high-risk investment losses. For example, our definition of a market turbulence period (21 days) refers to a turbulent regime lasting more than 21 consecutive days in the Bitcoin market; buying is prohibited during a turbulent regime that continues longer than 21 days (= Bitcoin direction changes, value converts to 0).

Table 11 shows the classification accuracy for the direction of Bitcoin prices in different market turbulence periods. It has been proven that the accuracy of the basic AdaBoost-LSTM model (= default) is further improved when it is combined with turbulence prediction information. In this study, the greatest increase in accuracy was achieved when the turbulence period was set between 21 and 28 days.

We, therefore, estimate that the longer the high fluctuation period, the greater the market uncertainty, and the lower the predictive power of the model. We further analyze the accuracy of each trading strategy in predicting the direction of Bitcoin prices according to volatility.

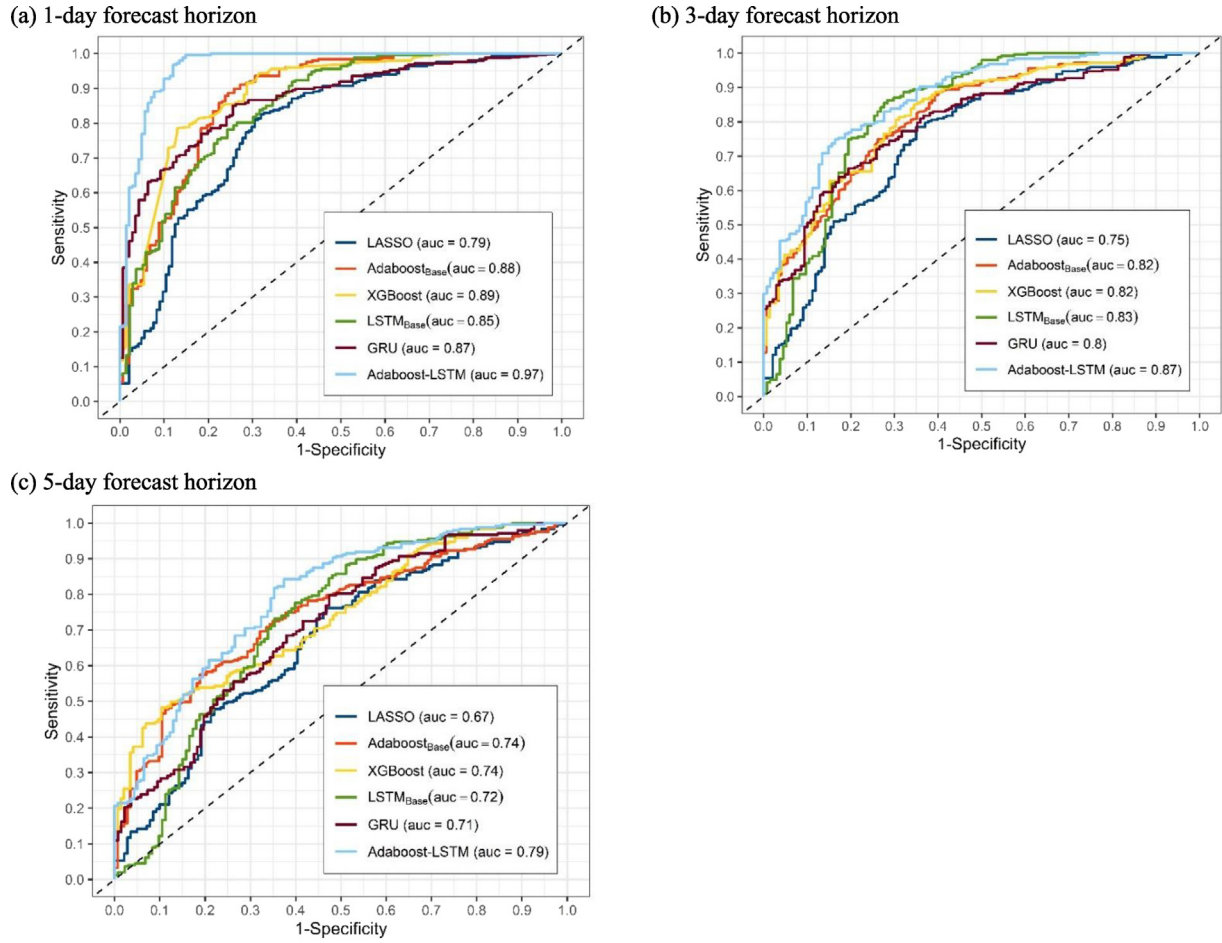


Fig. 8. ROC curve of classification models for turbulent regimes according to different forecast horizons: (a) 1-day, (b) 3-day, and (c) 5-day forecast horizons.

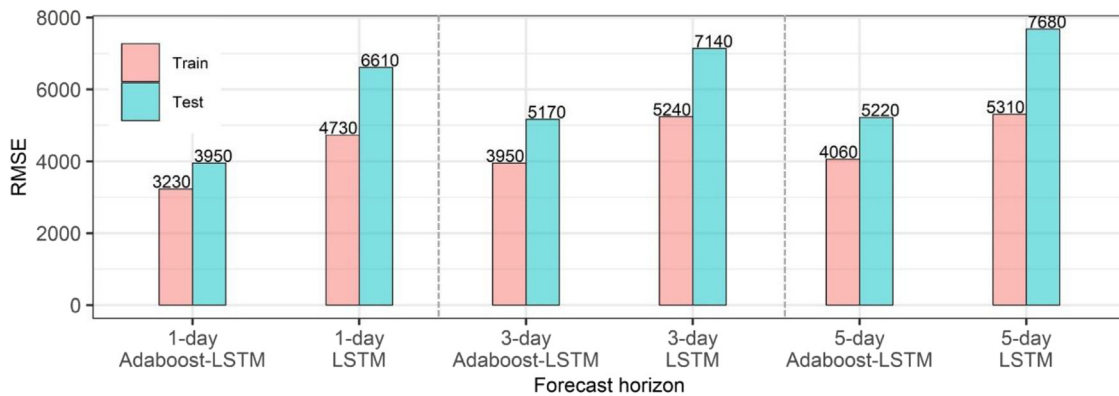


Fig. 9. Training and test errors of AdaBoost-LSTM model and LSTM_{base} for Bitcoin price prediction.

Table 12 shows the classification accuracy for the direction of Bitcoin prices of each trading strategy at different standard deviations of the logarithms of the prices. In Table 12, the accuracy of the AdaBoost-LSTM model is better than that of traditional trading strategies under high-volatility conditions of more than 1 sigma. In particular, fortified with market turbulence information, our model shows a 21.5sigma compared to the unfortified AdaBoost-LSTM model.

Next, in order to demonstrate the practicality of our trading strategy, we calculate daily profit and loss during the test set period according to the trading process based on Algorithm 2. The basic concept is that investors buy at the closing price when the state of the trading transaction changes to 'Buy,' and on the contrary, sell when the trading transaction changes to 'Sell.' If the transaction state does not change, 'Hold' the current portfolio position. The transaction fee of cryptocurrency varies slightly by

Table 10

Comparison of predictions of Bitcoin direction by each trading method according to different forecast horizons.

Forecast horizon	Trading method	Accuracy	Sensitivity	Specificity	Balanced accuracy
1 day	Long-only	0.480	1.000	–	0.497
	MACD	0.500	0.463	0.534	0.499
	ROC	0.528	0.484	0.569	0.526
	Stochastic	0.523	0.484	0.559	0.522
	AdaBoost-LSTM	0.541	0.768	0.328	0.548
3 days	Long-only	0.490	1.000	–	0.500
	MACD	0.508	0.469	0.545	0.507
	ROC	0.505	0.464	0.545	0.504
	Stochastic	0.526	0.484	0.565	0.525
	AdaBoost-LSTM	0.543	0.536	0.550	0.543
5 days	Long-only	0.482	1.000	–	0.500
	MACD	0.454	0.410	0.495	0.452
	ROC	0.492	0.452	0.530	0.491
	Stochastic	0.531	0.500	0.559	0.530
	AdaBoost-LSTM	0.577	0.527	0.624	0.575

Table 11

Comparison of predictions of Bitcoin direction according to different settings of the turbulent period. Default is the basic AdaBoost-LSTM model without turbulent information.

Forecast horizon	Turbulent period	Accuracy	Sensitivity	Specificity	Balanced accuracy
1 day	Default	0.541	0.768	0.328	0.548
	7 days	0.536	0.412	0.652	0.532
	14 days	0.544	0.545	0.542	0.544
	21 days	0.595	0.620	0.572	0.596
	28 days	0.582	0.610	0.557	0.583
3 days	Default	0.543	0.536	0.550	0.543
	7 days	0.528	0.278	0.768	0.523
	14 days	0.549	0.369	0.722	0.545
	21 days	0.572	0.428	0.711	0.570
	28 days	0.585	0.476	0.691	0.583
5 days	Default	0.577	0.527	0.624	0.575
	7 days	0.521	0.209	0.814	0.511
	14 days	0.536	0.262	0.794	0.528
	21 days	0.611	0.471	0.744	0.607
	28 days	0.604	0.465	0.734	0.599

Table 12Comparison of predictions of Bitcoin direction by each trading method according to variations in volatility during the test period. The volatility distribution includes values for all forecast horizons such as 1-day forecast horizon (mean μ : -0.105 , SD σ : 3.074), 3-day forecast horizon (μ : -0.308 , σ : 5.818), and 5-day forecast horizon (μ : -0.501 , σ : 7.647).

Log volatility (θ)	Share of total 100 (%)	Long-only	MACD	ROC	Stochastic oscillator	AdaBoost-LSTM	AdaBoost-LSTM with turbulence information
$\theta < \mu \pm 1\sigma $	68.3	0.479	0.482	0.508	0.525	0.547	0.555
$ \mu \pm 1\sigma \leq \theta < \mu \pm 2\sigma $	27.2	0.527	0.512	0.488	0.488	0.605	0.630
$ \mu \pm 2\sigma \leq \theta$	4.6	0.379	0.455	0.591	0.697	0.636	0.773

Table 13

Comparison of cumulative profit and Sharpe ratio by trading method during test period according to different forecast horizons. All indices are based on the end of the day.

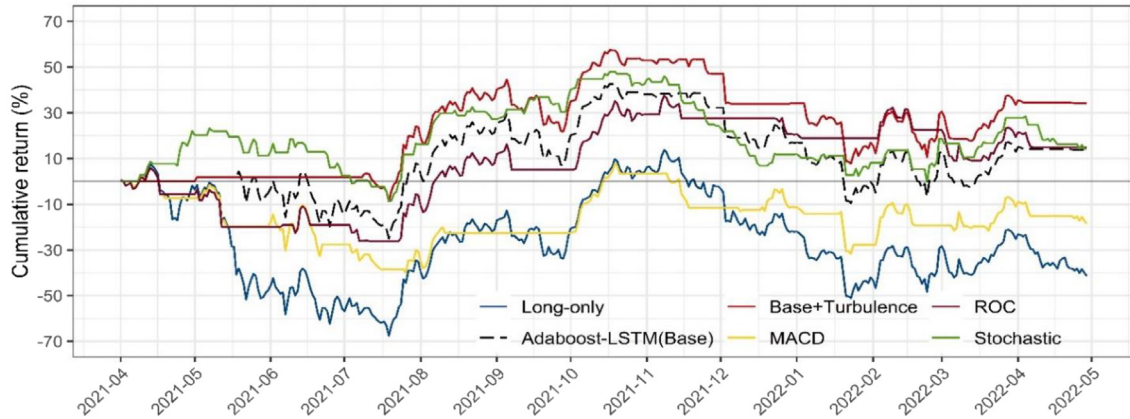
Forecast horizon	Metrics	Long-only	MACD	ROC	Stochastic slow	AdaBoost-LSTM	AdaBoost-LSTM with turbulence information
1 day	Cumulative profits (%)	−41.38	−18.56	14.91	14.28	13.86	34.26
	Sharpe ratio	−0.034	−0.021	0.021	0.014	0.016	0.044
3 days	Cumulative profits (%)	−134.65	−33.22	−13.45	9.28	64.70	116.8
	Sharpe ratio	−0.053	−0.049	−0.009	0.006	0.046	0.100
5 days	Cumulative profits (%)	−195.44	−88.85	−55.45	40.87	76.40	165.16
	Sharpe ratio	−0.066	−0.096	−0.065	0.020	0.036	0.091

the exchange between 0 and 0.25% [102]. We assumed that the transaction cost is fixed at 0.1% [103], and deduct the fee when the transaction state only changes.

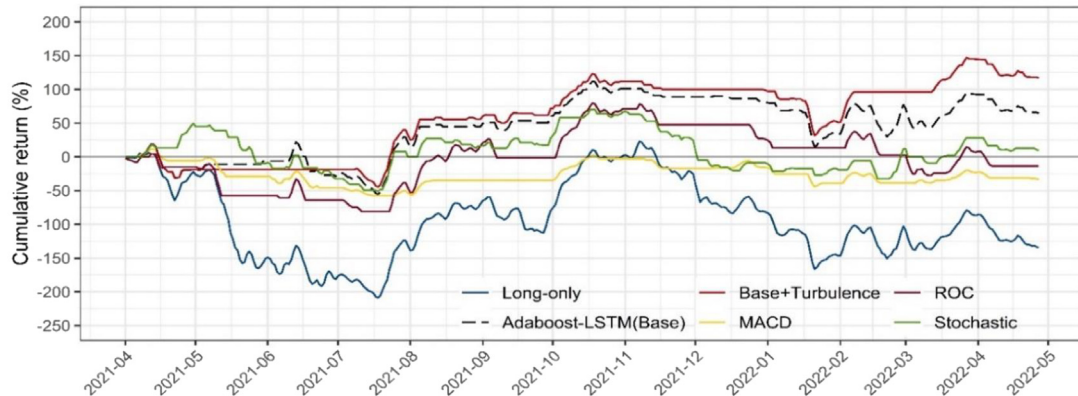
Fig. 10 plots the cumulative profits over time during the test set period for comparison with the other trading strategies, and Table 13 summarizes the cumulative profits and Sharpe ratio.

In Table 13, among traditional strategies, the stochastic oscillator was the best. However, for all forecast horizons, cumulative returns and the Sharpe ratio of our AdaBoost-LSTM model are 3.71 and 3.54 times higher on average, respectively, than those of the stochastic oscillator. In addition, our model outperformed other strategies in the medium term (3-day and 5-day forecast

(a) 1-day forecast horizon



(b) 3-day forecast horizon



(c) 5-day forecast horizon

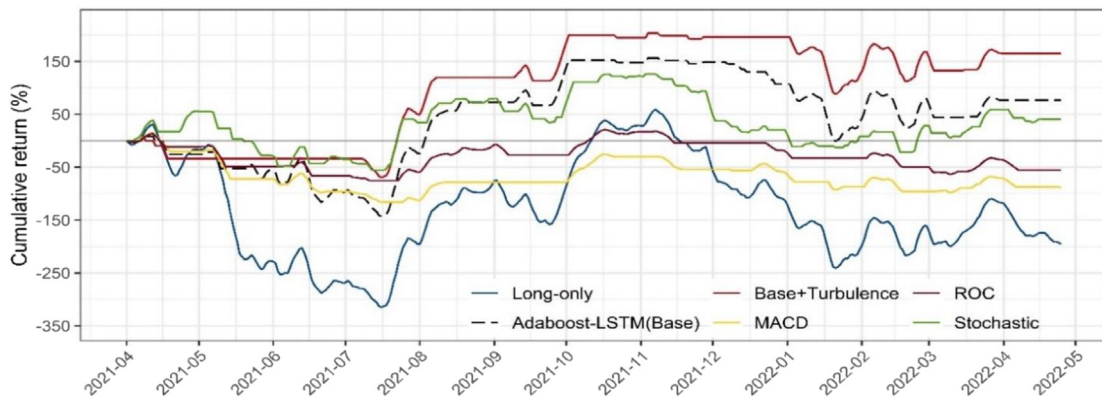


Fig. 10. Cumulative Bitcoin profits of each trading strategy according to different forecast horizons: (a) 1-day, (b) 3-day, and (c) 5-day.

horizons) more than in the short term (1-day forecast horizon). Results for the long-only strategy, with which no technical method was associated, are very poor in terms of profitability to the point of recording a negative cumulative return. Interestingly, adding market turbulence information to the model, especially information about turbulent regimes predicted to last longer than 21 days, further increases the cumulative returns of the AdaBoost-LSTM model.

Specifically, for all forecast horizons, AdaBoost-LSTM, with knowledge of market turbulence based on the MS model, yields cumulative returns 2.02 times higher and Sharpe ratios 2.48 times higher on average than those of the basic AdaBoost-LSTM model. However, when the period of market turbulence is defined tightly (within 14 days), cumulative returns are smaller than those with the basic AdaBoost-LSTM model. (Fig. E.4 in Appendix E provides a simulation of cumulative returns according to different periods

of market turbulence.)

Many studies have found that a period of high volatility can initially contribute to an increase in profits, but if market turbulence is prolonged, it can adversely affect earnings due to high uncertainty [26,37,104]. In this study, we conclude that the “no-buy” signal could play a major role in reducing losses during periods of prolonged turbulence, especially volatility predicted to last longer than 21 days.

Finally, we verified that our trading system had a statistically significant relationship with profits or the Sharpe ratio. Table 14 shows the OLS regression of the relationship between daily profit and the trading signal of each trading strategy. In general, the effect of traditional trading strategies on daily profit was not statistically significant, whereas the trading signals of our AdaBoost-LSTM model had a significant effect on daily profit. In addition, the AdaBoost-LSTM model with turbulence knowledge

Table 14

Comparison of OLS regression on the effect of daily profits(%) by trading method during the test period according to different forecast horizons.

Forecast horizon	OLS results	MACD	ROC	Stochastic slow	AdaBoost-LSTM	AdaBoost-LSTM with turbulence information
1 day	Coefficient	−0.04	0.04	0.03	0.04	0.10
	<i>t</i> -stat	−1.05	1.71*	2.31**	0.91	1.74*
	Standard error	(0.10)	(0.10)	(0.10)	(0.13)	(0.11)
3 days	Coefficient	−0.08	−0.03	0.02	0.19	0.21
	<i>t</i> -stat	−0.31	−1.31	0.71	3.31***	2.36**
	Standard error	(0.09)	(0.19)	(0.19)	(0.21)	(0.17)
5 days	Coefficient	−0.23	−0.14	0.1	0.21	0.46
	<i>t</i> -stat	−0.43	−0.03	1.93*	1.81*	5.23***
	Standard error	(0.12)	(0.11)	(0.26)	(0.29)	(0.26)

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

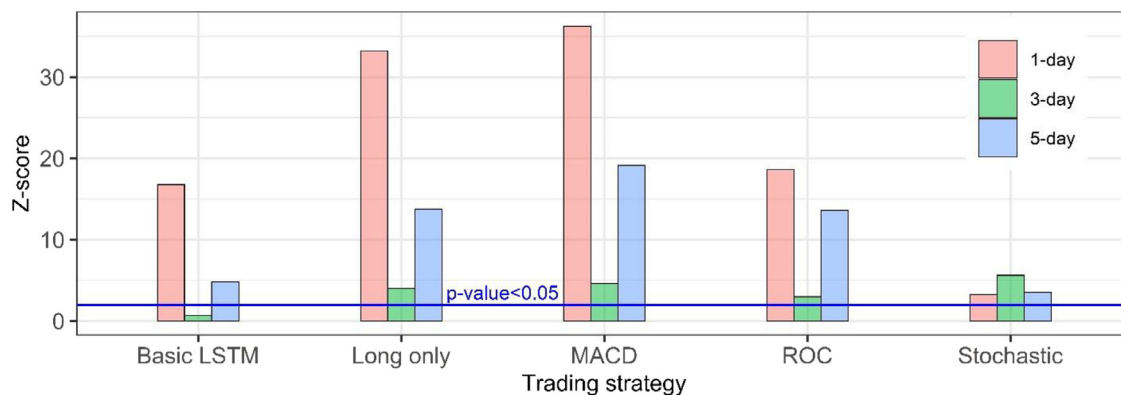


Fig. 11. Z-test for daily Sharpe ratio between AdaBoost-LSTM with turbulence information and other models. The daily Sharpe index is based on cumulative past to present. The blue line indicates the significance levels of 0.05 based on the two-sided test.

had a statistically significant relationship with daily profit in all forecast horizons. In particular, in midterm trading (3- and 5-day forecast), our daily trading signal significant increases in profit.

Fig. 11 shows the z-test verification for the daily Sharpe ratio between the AdaBoost-LSTM with turbulence knowledge and other trading strategies. The blue line illustrates that the mean difference in the Sharpe ratio between the two samples is statistically significant at the level of 0.05 in all forecast horizons.

5. Conclusion

In this study, we develop an intelligent Bitcoin trading system using an ensemble prediction method that integrates the AdaBoost framework and the LSTM DL. First, through our novel model fusion approach, predictive performance was greatly improved by combining the strengths of these two models, such as the high-representational capacity of LSTM and the overfitting minimization of AdaBoost. Second, our study further combines economic knowledge of market turbulence obtained via Markov modeling with the AdaBoost-LSTM prediction model, thus helping to avoid high risk due to volatility, a chronic problem related to Bitcoin investment. In particular, we found that a turbulent market regime adversely affects Bitcoin profits with time lags, especially when the turbulent regime is prolonged (21 consecutive days or more), which is a high-risk period.

Finally, we implement an intelligent trading system to maximize profits by predicting upward movement in the Bitcoin market and reducing loss by warning against trade at times of high risk due to turbulence. The simulation of our trading system for one year (from April 1, 2021 to April 30, 2022) showed cumulative returns and Sharpe ratios that greatly exceeded those of other trading strategies such as the stochastic oscillator and

MACD. Results were especially excellent in the medium term (3- or 5-day forecast horizons) versus trading strategies in the short term (1-day forecast horizon).

Currently, U.S. monetary policy is rapidly shifting towards austerity in order to suppress extremely high inflation caused by COVID-19. Accordingly, in the rapidly changing global financial system, uncertainty and volatility in the cryptocurrency market continue to expand. Since asset markets such as stocks and bonds are interconnected, the instability of the global financial system can cause a large financial shock to all asset markets, including cryptocurrency [105]. Therefore, it is important to minimize high risks in the future to encourage cryptocurrency investment. Our intelligent trading system based on the sequential fusing of various frameworks provides an efficient investment strategy that is both low-risk and promises high returns to investors. However, there is a concern that policy regulations related to cryptocurrency will be strengthened in the future and that the issuance of CBDC (Central Bank Digital Currency) may cause unpredictable volatility that even our methodology may not offset.

CRedit authorship contribution statement

Sangjin Park: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization. **Jae-Suk Yang:** Conceptualization, Methodology, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Tuning procedure of the LSTM model

In our study, a lot of hyperparameter tuning tasks are required to construct an LSTM model; we performed a sequential grid search to find good hyperparameter combinations. The overall tuning process consists of three stages, and conceptually, the optimized parameters in the previous stage are employed sequentially in the next stage.

In the first stage, we optimized the main configuration of the number of neurons in a single layer from 1 and 100 during the learning process based on the default setting⁴. Fig. A.1(a) shows that the prediction error is the lowest when 50 nodes, but as the number of nodes exceeds 30, there is no significant difference in prediction error.

For testing multi-layers, we set up each LSTM layer containing 50 hidden neurons chosen by the previous step, and Fig. A.1(b) shows that the setting of two LSTM layers has the best performance.

In the second stage, minor configurations are verified. Fig. A.2 plots the process of searching optimized points of parameters one by one.

The prediction error was the smallest under the following conditions: (1) epochs: 150, (2) learning rate: 0.001, (3) dropout rate: 0.2, and (4) Optimizer: Adam. In addition, mean square error and the cross-entropy function were used as the loss function and the Relu function as the activation function.

In the third, most important stage, we set a timestep that determines how long to preserve historical time-series from each time t. That is, it determines at what level the observation length for the cycle will be reflected in the prediction model. Fig. A.3 tested the best-performing timesteps from the 1-day to the 10 days and showed that the error of prediction is lowest when the timestep is 5 days in the 1- and 5-day forecast model and 7 days in the 3-day forecast model.

(1) First stage: See Fig. A.1

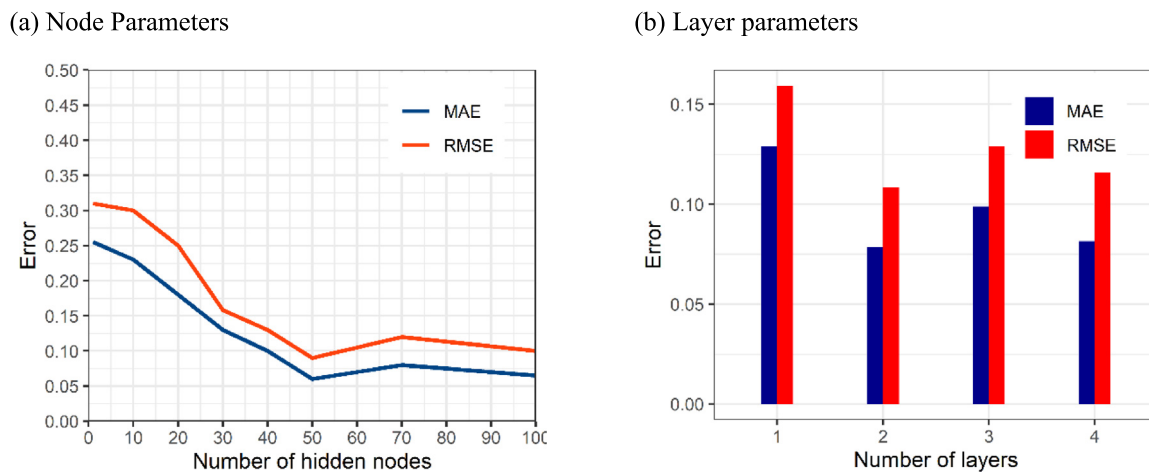
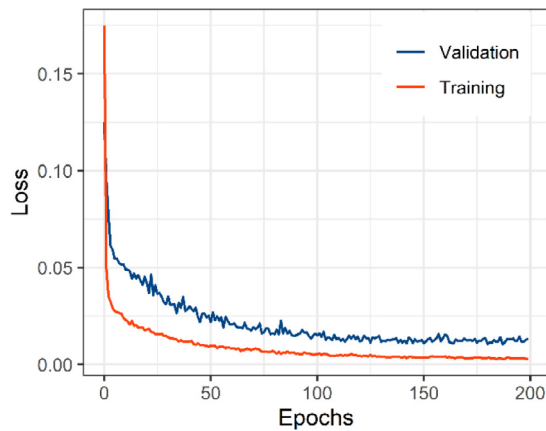


Fig. A.1. First configuration: prediction error of Bitcoin price forecast model according to changes in (a) hidden nodes and (b) layer parameters. All models are based on a 1-day forecast horizon.

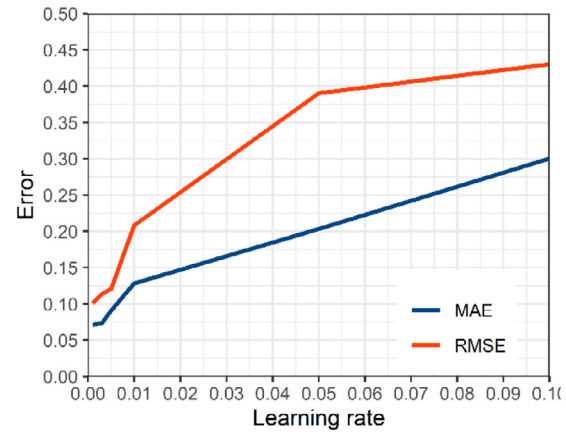
(2) Second stage: See Fig. A.2

⁴ Batch size: 30, Dropout rate: 0.001, Epochs: 100, and Learning rate: 0.001, Optimizer: Adam, and Timestep: 5 days.

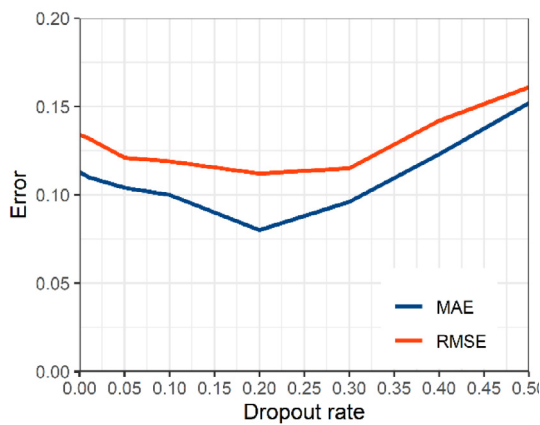
(a) Epochs parameters



(b) Learning rate



(c) Dropout rate parameter



(d) Optimizer

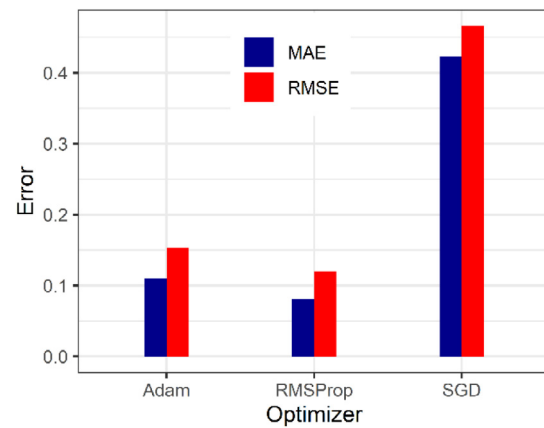


Fig. A.2. Second configuration: prediction error of Bitcoin price forecast model according to changes in (a) Epochs, (b) Learning rate, (c) Dropout rate, (d) Optimizer. All models are based on a 1-day forecast horizon.

(3) Third stage: See Fig. A.3

(a) Timestep

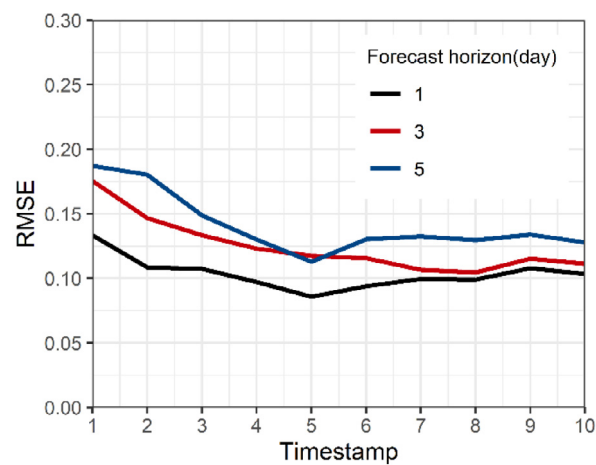


Fig. A.3. Third configuration: prediction error of Bitcoin price forecast model according to changes in timesteps. The model is set using different forecast horizons such as 1-day, 3-days, and 5-days. The x-axis is timestep, the unit of time is the day, and the y-axis is the error of prediction measured using RMSE indicators.

Therefore, in our study, the basic structure of our LSTM model is followings:

- The input layer had 16 features, and the LSTM layer consisted of two layers containing 50 nodes and 30 nodes.
- The dense layer consisted of three layers containing, two of which have 30 nodes, and the output layer has 1 node. The output layer uses sigmoid activation functions.
- The batch size is 30, and the number of epochs for training is 150.
- The learning rate and the dropout rate were 0.001 and 0.2, respectively.
- The timesteps were set according to the forecast period (e.g., timesteps 5 days for the 1- and 5-day forecasting model and 7 days set for the 3-day forecasting model).
- We used the Adam optimizer, the Relu activation function.

In addition, we optimize hyperparameters for other baseline models such as LASSO, AdaBoost_{base} and XGBoost through random search. However, since the GRU model has a structure similar to the LSTM model, it was set to the same hyperparameters of the LSMT model.

Model	Hyperparameters	Search space	Selected value
LASSO	C	range(0.001, 0.1, by = 0.001)	0.0057
AdaBoost _{base}	Learning rate	[0.001, 0.01, 0.1, 0.5, 1.0, 1.5]	0.01
	Number of weak learners	[50, 100, 200, 500, 1000, 2000]	200
	Min samples leaf	[3, 5, 7, 10, 15]	6
	Max depth	[3, 5, 7, 10, 20, 30]	20
	Max leaf nodes	[3, 5, 10, 15, 20, 30]	30
XGBoost	Number of weak learners	[10, 50, 100, 150, 200, 300]	200
	Learning rate	[0.001, 0.01, 0.1, 0.5, 1.0, 1.5]	0.01
	Max_depth	[5, 10, 15, 20, 30]	10
	Subsample	[0.1, 0.3, 0.5, 0.7, 1.0]	0.7
	Min_child_weight	[1, 2, 3, 4, 5, 6]	1

Note: Optimal parameters for each predictive model are selected via random search. All other conditions are default parameters. AdaBoost_{base} is a basic ensemble model using a decision tree as the weak learner.

Appendix B. Turbulent regime periods

B.1. Description of start-end points, returns, and duration during prolonged periods of turbulence lasting more than 7 consecutive days.

See [Table B.1](#)

Table B.1
Detailed summary for periods of turbulent regime.

Number	Type	Start day	End day	Daily log return (%)		Duration
				Average	Cumulative profit	
1	Train set	2016-06-12	2016-07-02	0.77	16.25	21
2		2016-12-21	2017-01-17	0.36	10.05	28
3		2017-03-07	2017-03-29	-0.91	-20.95	23
4		2017-05-01	2017-06-20	1.36	69.53	51
5		2017-06-23	2017-07-04	-0.32	-3.79	12
6		2017-07-09	2017-10-02	0.64	55.04	86
7		2017-10-07	2017-11-20	1.37	61.71	45
8		2017-11-25	2018-04-14	-0.1	-1.42	141
9		2018-04-20	2018-05-24	-0.24	-8.48	35
10		2018-06-22	2018-06-30	-0.59	-5.35	9
11		2018-07-22	2018-08-15	-0.59	-15.22	25
12		2018-11-14	2018-12-30	-1.07	-50.42	47
13		2019-01-09	2019-01-16	-1.3	-10.42	8
14		2019-05-03	2019-06-04	1.13	37.41	33
15		2019-06-14	2019-08-05	0.67	35.69	53

(continued on next page)

Table B.1 (continued).

Number	Type	Start day	End day	Daily log return (%)		Duration
				Average	Cumulative profit	
16	Valid set	2019-08-13	2019-08-29	−1.04	−17.64	17
17		2019-09-23	2019-09-29	−3.01	−21.04	7
18		2019-11-19	2019-12-01	−0.99	−12.93	13
19		2020-01-07	2020-01-15	1.63	14.7	9
20		2020-02-18	2020-02-27	−0.95	−9.53	10
21		2020-03-08	2020-04-13	−0.79	−29.26	37
22		2020-04-28	2020-05-15	1.18	29.99	18
23		2020-07-26	2020-08-02	2.02	16.19	8
24		2020-11-16	2020-12-01	1.12	16.76	15
25		2020-12-09	2020-12-18	1.89	18.91	10
26		2020-12-24	2021-03-27	0.91	85.28	94
27		2021-04-16	2021-07-08	−0.77	−64.73	84
28		2021-07-19	2021-08-08	1.57	33.01	21
29	Test set	2021-09-19	2021-10-07	0.63	11.91	19
30		2021-10-19	2021-10-27	−0.41	−3.73	9
31		2021-11-16	2021-12-29	−0.71	−31.35	44
32		2022-01-21	2022-02-08	0.20	3.76	19
33		2022-02-15	2022-03-11	−0.30	−7.5	25
34		2022-04-06	2022-04-12	−2.03	−14.23	7

Appendix C. Performance of AdaBoost-LSTM model according to different numbers N of LSTM learners

C.1. Comparison of prediction error for bitcoin prices according to different forecast horizons

Forecast horizon	N	MAE	RMSE	MAPE	R^2
1 day	0	5480	6610	11.285	0.420
	2	4920	5730	10.220	0.564
	3	4430	5250	10.075	0.634
	5	4270	5080	9.170	0.679
	10	4230	5210	9.234	0.647
	20	4050	5030	8.653	0.670
	30	3940	4840	8.196	0.696
	40	3260	3950	7.010	0.797
	50	3530	4260	8.017	0.730
3 days	0	5740	7140	11.563	0.332
	2	5110	6380	10.478	0.450
	3	4980	6040	10.412	0.540
	5	5200	6490	11.520	0.432
	10	4470	5670	9.458	0.566
	20	4690	5530	10.519	0.587
	30	4150	5170	8.754	0.650
	40	4280	5430	8.810	0.614
	50	4460	5490	9.345	0.606
5 days	0	6110	7680	12.256	0.199
	2	5860	7110	12.889	0.314
	3	5620	6930	11.896	0.343
	5	5290	6660	11.000	0.397
	10	5640	6750	11.852	0.398
	20	5090	6150	10.812	0.501
	30	4660	5550	10.438	0.595
	40	4320	5220	9.393	0.645
	50	4330	5490	9.026	0.603

C.2. Comparison of classification accuracy for turbulent regimes in bitcoin market

Forecast horizon	N	Accuracy	Sensitivity	Specificity	Balanced accuracy	AUC
1 day	0	0.789	0.776	0.816	0.796	0.852
	2	0.807	0.729	0.949	0.839	0.905
	3	0.880	0.846	0.941	0.894	0.929
	5	0.916	0.903	0.941	0.922	0.949
	10	0.812	0.741	0.941	0.841	0.943
	20	0.894	0.862	0.950	0.906	0.966
	30	0.845	0.785	0.950	0.868	0.951
	40	0.930	0.996	0.816	0.906	0.966
	50	0.892	0.875	0.922	0.898	0.959
3 days	0	0.778	0.909	0.537	0.723	0.825
	2	0.801	0.895	0.627	0.761	0.840
	3	0.819	0.943	0.590	0.766	0.838
	5	0.819	0.943	0.590	0.766	0.840
	10	0.808	0.923	0.597	0.760	0.840
	20	0.819	0.931	0.612	0.772	0.844
	30	0.816	0.901	0.647	0.774	0.870
	40	0.806	0.887	0.657	0.772	0.853
	50	0.806	0.911	0.612	0.761	0.849
5 days	0	0.712	0.789	0.571	0.680	0.722
	2	0.732	0.858	0.504	0.681	0.744
	3	0.740	0.875	0.496	0.685	0.740
	5	0.740	0.891	0.467	0.679	0.744
	10	0.734	0.850	0.526	0.688	0.743
	20	0.742	0.907	0.445	0.676	0.755
	30	0.767	0.846	0.626	0.736	0.783
	40	0.750	0.826	0.613	0.720	0.772
	50	0.759	0.875	0.554	0.714	0.789

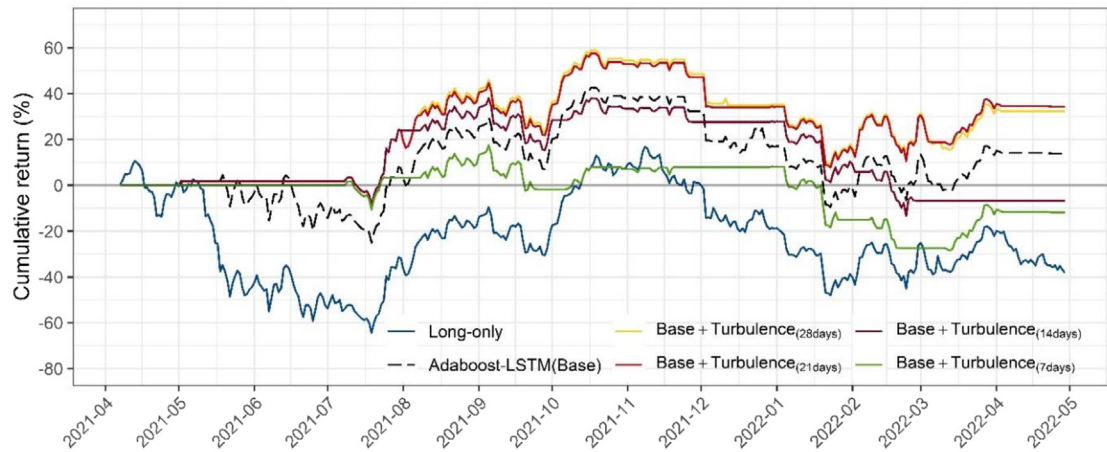
Appendix D. Description of traditional trading indicators

Indicators	Description	Parameters	Formula	Condition of “buy” transaction
SMA (Simple Moving Average)	Average prices over selected period of time	SMA(n-days)	$\frac{1}{n} \sum_{t=1}^n Price_t$	–
MACD (Moving Average Convergence Divergence)	Trend momentum for relationship between two moving averages (short and medium term)	MACD(12, 26, 9)	MACD = SMA(12) – SMA(26) Signal = SMA(MACD, 9)	If MACD > Signal
ROC (Rate-of-Change)	Percentage change in price over selected period of time	ROC(14)	$\frac{Price_t - Price_{t-14}}{Price_{t-14}} \times 100$	If ROC > 0
Stochastic oscillator	Location of the price relative to the highest–lowest range over selected period of time	Stochastic Fast%K(5) Stochastic Fast%D(3) Stochastic Slow%K(3) Stochastic Slow%D(3)	$\frac{Price_t - MinPrice(5days)}{MaxPrice(5days) - MinPrice(5days)} \times 100$ SMA(3) of Stochastic Fast %K(5) SMA(3) of Stochastic Fast %D(3) SMA(3) of Stochastic Slow %K(3)	If Stochastic Slow %K > Stochastic Slow %D

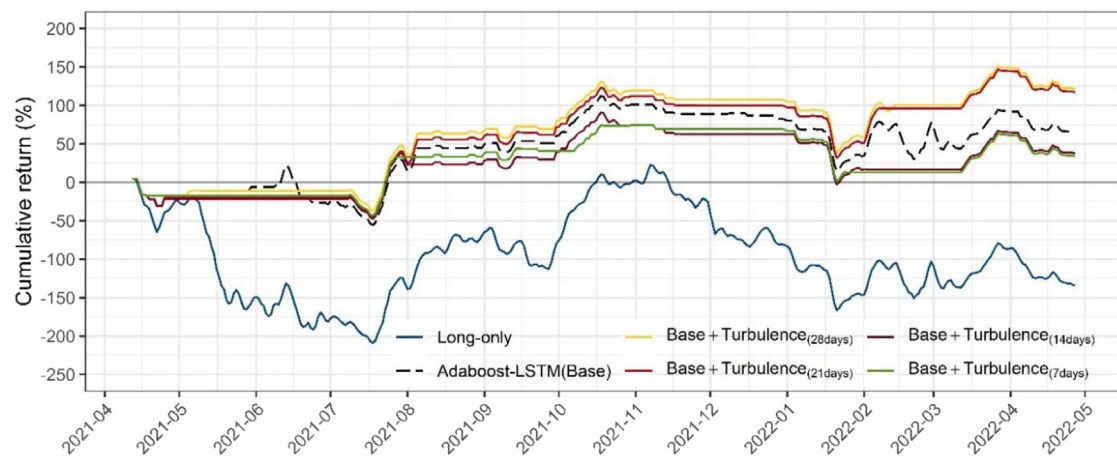
Appendix E. Cumulative bitcoin profits according to different periods of market turbulence

See Fig. E.4.

a) 1-day forecast horizon



(b) 3-day forecast horizon



(c) 5-day forecast horizon

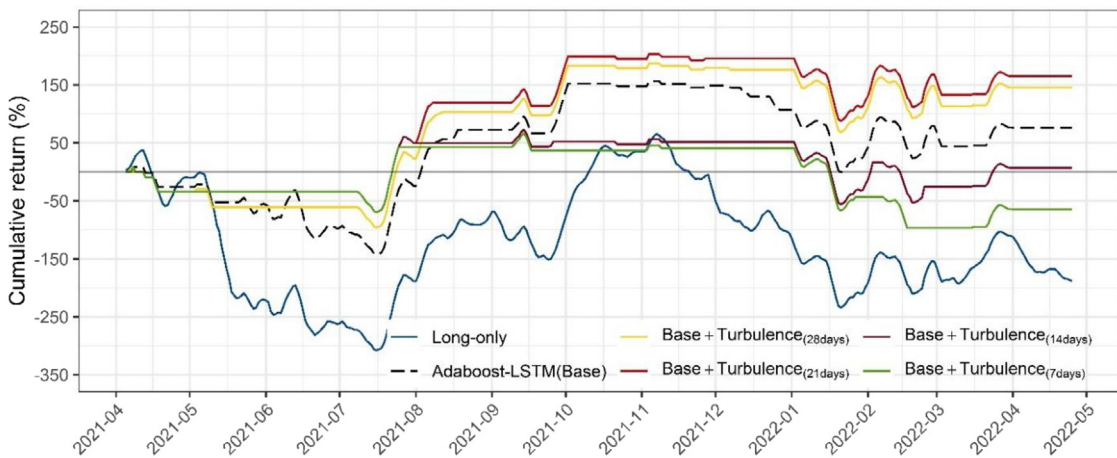


Fig. E.4. Cumulative Bitcoin profits based on different forecast horizons: (a) 1-day, (b) 3-day, and (c) 5-day. The turbulence period (in parentheses) indicates a continuously prolonged period of the turbulent regime.

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